ResearchGate

See discussions, stats, and author profiles for this publication at: https://www.researchgate.net/publication/378523388

Comparison of conventional and machine learning methods for bias correcting CMIP6 rainfall and temperature in Nigeria

Article *in* Theoretical and Applied Climatology · February 2024

DOI: 10.1007/s00704-024-04888-9

CITATIONS
11

READS
430

8 authors, including:



Bashir Tanimu
University of Technology Malaysia
13 PUBLICATIONS **30** CITATIONS
SEE PROFILE



Al-Amin Danladi Bello
Ahmadu Bello University
23 PUBLICATIONS **120** CITATIONS
SEE PROFILE



Sule Argungu Abdullahi
Ahmadu Bello University
26 PUBLICATIONS **117** CITATIONS
SEE PROFILE



Morufu Ajibike
Ahmadu Bello University
10 PUBLICATIONS **63** CITATIONS
SEE PROFILE

All content following this page was uploaded by Mohammad Kamruzzaman on 19 June 2025.

The user has requested enhancement of the downloaded file.

Theoretical and Applied Climatology
https://doi.org/10.1007/s00704-024-04888-9

RESEARCH

Comparison of conventional and machine learning methods for bias correcting CMIP6 rainfall and temperature in Nigeria

Bashir Tanimu[1,2] · Al-Amin Danladi Bello[1] · Sule Argungu Abdullahi[1] · Morufu A. Ajibike[1] · Zaher Mundher Yaseen[3,5] · Mohammad Kamruzzaman[4] · Mohd Khairul Idlan bin Muhammad[2] · Shamsuddin Shahid[2,6]

Received: 3 November 2023 / Accepted: 16 February 2024
© The Author(s), under exclusive licence to Springer-Verlag GmbH Austria, part of Springer Nature 2024

Abstract

This research assesses the efficacy of thirteen bias correction methods, including traditional and machine learning-based approaches, in downscaling four chosen GCMs of Coupled Model Intercomparison Project 6 (CMIP6) in Nigeria. The 0.5° resolution gridded rainfall, maximum temperature (Tmx), and minimum temperature (Tmn) of the Climate Research Unit (CRU) for the period 1975 – 2014 was used as the reference. The Compromise Programming Index (CPI) was used to assess the performance of bias correction methods based on three statistical metrics. The optimal bias-correction technique was employed to rectify bias to project the spatiotemporal variations in rainfall, Tmx, and Tmn over Nigeria for two distinct future timeframes: the near future (2021–2059) and the distant future (2060–2099). The study's findings indicate that the Random Forest (RF) machine learning technique better corrects the bias of all three climate variables for the chosen GCMs. The CPI of RF for rainfall, Tmx, and Tmn were 0.62, 0.0, and 0.0, followed by the Power Transformation approach with CPI of 0.74, 0.36, and 0.29, respectively. The geographic distribution of rainfall and temperatures significantly improved compared to the original GCMs using RF. The mean bias-corrected projections from the multimodel ensemble of the GCMs indicated a rainfall increase in the near future, particularly in the north by 2.7–12.7%, while a reduction in the south in the far future by -3.3% to -10% for different SSPs. The temperature projections indicated a rise in the Tmx and Tm from 0.71 °C and 0.63 °C for SSP126 to 2.71 °C and 3.13 °C for SSP585. This work highlights the significance of comparing bias correction approaches to determine the most suitable approach for adjusting biases in GCM estimations for climate change research.

✉ Shamsuddin Shahid
sshahid@utm.my

Bashir Tanimu
bashirtanimu10@gmail.com

Al-Amin Danladi Bello
ask4alamin@gmail.com

Sule Argungu Abdullahi
abdummu1@gmail.com

Morufu A. Ajibike
ajibikefarida2012@gmail.com

Zaher Mundher Yaseen
z.yaseen@kfupm.edu.sa

Mohammad Kamruzzaman
milonbrri@gmail.com

Mohd Khairul Idlan bin Muhammad
mohdkhairulidlan@utm.my

[1] Department of Water Resources and Environmental Engineering, Ahmadu Bello University, Zaria, Nigeria

[2] Faculty of Civil Engineering, Universiti Teknologi Malaysia (UTM), 81310 Johor Bahru, Malaysia

[3] Civil and Environmental Engineering Department, King Fahd University of Petroleum & Minerals, 31261 Dhahran, Saudi Arabia

[4] Farm Machinery and Postharvest Technology Division, Bangladesh Rice Research Institute, Gazipur 1701, Bangladesh

[5] Interdisciplinary Research Centre for Membranes and Water Security, King Fahd University of Petroleum & Minerals, 31261 Dhahran, Saudi Arabia

[6] Environmental and Atmospheric Sciences Research Group, Scientific Research Center, Al-Ayen University, 64001 Thi-Qar, Nasiriyah,, Iraq

Published online: 27 February 2024

1 Introduction

Global climate models (GCMs) are widely regarded as indispensable tools in climate change research due to their ability to project future climate change and accurately simulate past climate patterns.(Gebresellase et al. 2022; Hartmann 2016; Shrestha & Pradhanang 2022). Evaluating climate change and supporting decision-making in mitigating climate change impacts are crucial (da Silva et al. 2022; Illangasingha et al. 2023). Numerous climate investigations have been undertaken using the Coupled Model Intercomparison Project Phase 6 (CMIP6) GCM simulation data (Peng et al. 2022; Wu et al. 2022). Nevertheless, it has been observed that GCMs exhibit limitations in terms of their temporal and spatial resolutions, resulting in inherent inaccuracies (Riahi et al. 2017; Zhang et al. 2022). Consequently, the use of GCM simulations for impact climate studies at a regional level is deemed unsuitable (Chokkavarapu & Mandla 2019; Jose et al. 2022). The development of downscaling and bias correction (BC) approaches aims to address the geographical disparity between the outputs of GCMs and the needed fine-scale data (Maraun & Widmann 2018; Nashwan et al. 2020).

Methods for downscaling include statistical and dynamical techniques. Dynamic approaches utilize GCMs as a fundamental tool for modeling physical processes and boundary conditions. These simulations are crucial for generating localized climate information at a regional scale (Iseri et al. 2021; Xu et al. 2019). Statistical downscaling involves establishing a correlation between the large-scale factors simulated by GCMs and the local climate patterns to predict local climate variables (Vogel et al. 2023). Statistical downscaling techniques have been more prevalent in recent years, mostly due to their advantageous characteristics, such as computing efficiency, simplicity, and ability to provide localized climatic data at specified geographical locations (Su et al. 2013; Zhang et al. 2022). Nevertheless, downscaled GCM often display notable biases, as highlighted (Senatore et al. 2022). It is necessary to implement BC algorithms to resolve this problem (Teutschbein & Seibert 2012).

BC involves adjusting GCM outputs to correct systematic inaccuracies, enhancing their alignment with observed data (Maraun 2016; Worako et al. 2022). Addressing bias in climate projections is crucial to prevent potentially misleading decision-making based on inaccurate outcomes (Maraun et al. 2017). Several BC techniques have been developed to address various forms of bias in GCMs effectively. These techniques include the Delta method (Haylock et al. 2006), Quantile mapping (Maraun 2013), Empirical cumulative distribution function (ECDF) (Gudmundsson et al. 2012), Power transformation (PT) (Vrac et al. 2012), Scaling method (SCLM) (Maraun et al. 2010), and others. Conventional statistical techniques are employed in the development of those methods.

Several studies have conducted evaluations and comparisons of various bias correction methods to correct GCM rainfall and temperature data (Homsi et al. 2020; Nashwan et al. 2020; Salman et al. 2022; Shiru & Park 2020; Tumsa 2022; Vogel et al. 2023(Nashwan et al. 2020; Salman et al. 2022; Shiru and Park 2020; Tumsa 2022; Vogel et al. 2023; Homsi et al. 2020). One such research conducted by Homsi et al. (2020) examined the efficacy of several bias correction approaches in rectifying precipitation data in Syria. The findings indicated that linear scaling was the most effective approach. Jaiswal et al. (2022) utilized three statistical bias correction approaches to rectify monsoon rainfall simulation in several agro-climatic zones in India. Jose & Dwarakish (2022) evaluated six different bias correction strategies. This evaluation aimed to assess the effectiveness of these techniques in correcting high-resolution daily temperature simulations in the Netravati basin, a tropical river basin located in India. Laux et al. (2021) evaluated four different BC strategies to analyze precipitation and temperature data. The researchers concluded that it is advisable to utilize an ensemble of these techniques in generating climate projections to conduct assessments on the impacts of climate change. According to Tan et al. (2020), hybrid Bayesian calibration strategies demonstrated superior performance compared to individual methods in the simulation of precipitation and streamflow. Wörner et al. (2019) conducted a comprehensive analysis to compare six different BC procedures to anticipate the potential effects of climate change on high flows. The findings of this study indicated that none of the examined strategies had a clear superiority over the others. Luo et al. (2018) have also demonstrated that no universally applicable BC method can be effectively implemented in all regions. In order to address the geographical and temporal variability of bias and the method-specific assumptions associated with BC, it is necessary to do a comparative analysis of several approaches (Worako et al. 2022).

Machine learning (ML) has a notable advantage in uncovering meaningful insights within the climate system through pattern identification and feature extraction methodologies. Hence, the utilization of ML in BC has experienced significant attention in recent times. (Cho et al. 2020; Hieronymus 2023; H. Kim et al. 2021; Ling et al. 2022). However, it has been observed that ML algorithms can lead to inaccurate estimation due to overfitting during training (Seo & Ahn 2023). Therefore, the improvement of BC using ML algorithms is not guaranteed. It is crucial to compare ML models with conventional statistical models to rectify the bias in GCMs.

Nevertheless, research on the comparative evaluation of machine learning and traditional statistical techniques in rectifying the bias of GCM is scarce. A recent study by Seo & Ahn (2023) compared QM with the ML model for correcting daily rainfall. This work endeavors to fill this void by conducting a comparative analysis of the efficacy of various ML techniques

and conventional BC procedures. However, a comprehensive study that compares the performance of all the widely used conventional and ML BC methods is still lacking.

Nigeria, like many developing countries, is particularly vulnerable to the impacts of climate change (Shiru et al. 2020; Wada et al. 2023). Biased GCM projections can lead to underestimation of the true risks, jeopardizing effective adaptation and mitigation strategies. Bias correction techniques can provide finer-grained information specific to Nigeria's diverse landscapes and ecosystems. This localized data can help tailor adaptation strategies to the specific needs of different regions and communities (Hassan et al. 2020; Madaki et al. 2023). It enables better-informed decision-making for critical issues like water resource management, agricultural planning, and disaster preparedness.

This research examines the performance of three ML BC and ten traditional BC approaches in correcting the biases in GCM rainfall, Tmx, and Tmn in Nigeria. Four GCMs that have been previously identified as appropriate for simulating the climatic conditions in Nigeria were utilized. The reference dataset utilized in this study was the rainfall and temperature data from the climatic research unit (CRU) from 1975 to 2014. The optimal approach was employed to rectify bias in subsequent estimates. The downscaled projections of the four GCMs were utilized to estimate the climate of Nigeria for four distinct scenarios (SS126, SSP245, SSP370, and SSP585), employing the multimodel ensemble (MME) mean.

2 Research site and data collection

2.1 Research site

Nigeria, located in the Atlantic Ocean, covers an area of 923,769 km^2 (Fig. 1) and is home to an estimated 230 million people. (Solarin et al. 2021). The area's geographical location is between the Sahel region in the north and the Gulf of Guinea in the south. According to Falola & Heaton (2008), Nigeria's geographical location may be identified on the map using the coordinates of 20–15.5°E for longitude and 20–15°N for latitude. Nigeria has a tropical climate and is characterized by six distinct agroecological zones: semi-arid area, mangroves, Guinea savanna, rainforests, Sudan savanna, and swamp forest (Adeaga et al. 2022). Nigeria shares its northern border with Niger, its northeastern border with Chad, its eastern border with Cameroon, and its western border with Benin (Shiru et al. 2019). The duration of various seasons varies depending on the precise geographical region. According to Oguntunde & Abiodun (2013),

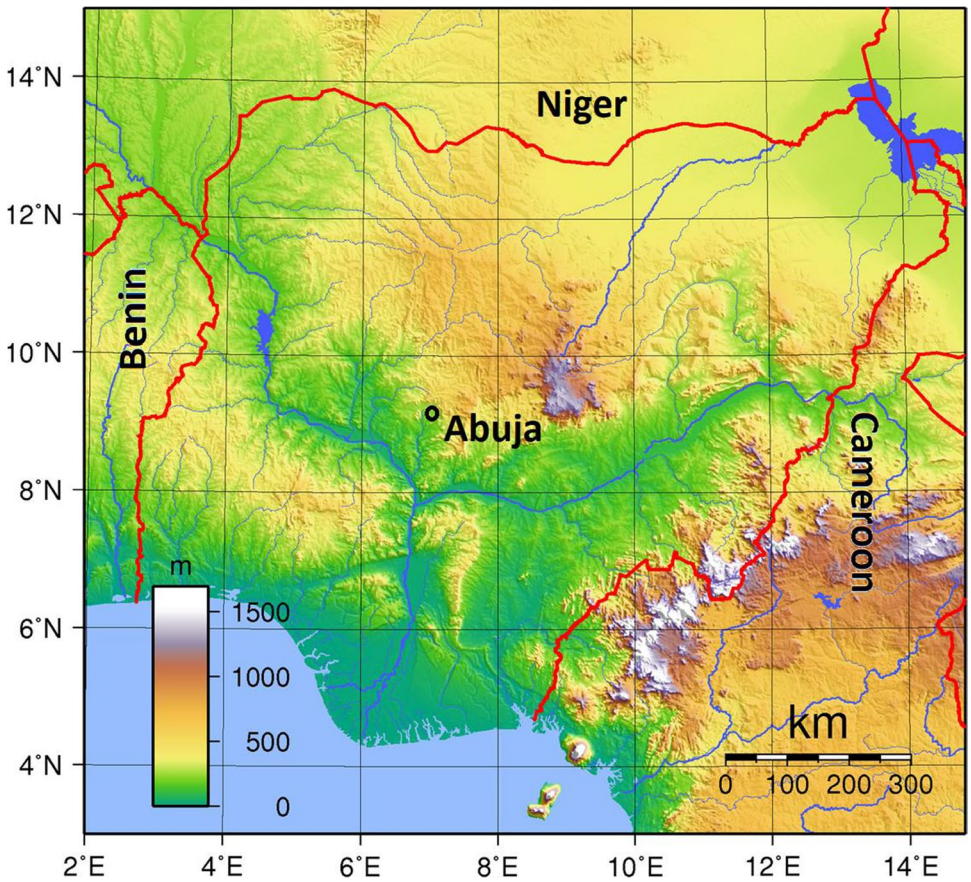


Fig. 1 The map of the study area

the rainy season in southern Nigeria spans from March to November, but in the northern area, it takes place from May to September. It is worth noting that the southern region has a greater rainfall duration than the northern region. Nigeria has significant disparities in yearly precipitation, with values ranging from around 500 mm/year in the semi-arid region of the north to 3000 mm/year in the mangrove and rainforest biological zones of the deep south (Enyidi 2017; Madaki et al. 2023). The humid climate in the region can be attributed to the close vicinity of the Gulf of Guinea (Ayanlade et al. 2019). According to Balogun et al. (2019), the northern areas of Nigeria are characterized by a prevailing dry season, during which temperatures can reach an average of 38 °C every month. In contrast, the southern regions maintain a mean temperature of around 32–33 °C. The duration of the harmattan wind, characterized by its arid and warm characteristics, has a lengthier persistence in the northern parts of Nigeria in comparison to the southern areas (Audu et al. 2013; Mobolade & Pourvahidi 2020).

A substantial portion of the Nigerian population relies on agriculture and is dependent on rainfall. The imperative to enhance food production in response to population increase poses considerable difficulties within the context of a dynamic climate (Shiru et al. 2020). Also, vegetation changes mostly result from precipitation and temperature fluctuations (Ahmad et al. 2023). The increase in flood occurrence is commonly ascribed to climate change, leading to significant consequences such as the loss of both human lives and property and the potential emergence of prolonged periods of drought in certain regions (Isa et al. 2023). Hence, establishing a dependable climate prediction holds significant importance for the nation. Nevertheless, the existing body of research on enhancing climate downscaling techniques in Nigeria remains limited (Shiru & Park 2020; Hassan et al. 2020; Wada et al. 2023; Isa et al. 2023).

2.2 Reference gridded data

In Nigeria, due to data shortage, inadequate data collection devices, and degraded weather and hydrological stations (Shiru et al. 2020; Wada et al. 2023), climate research unit (CRU) monthly rainfall, Tmx and Tmn datasets were used as the reference point for the historical period between 1975 to 2014. The CRU datasets are derived from the CRU version 4.01 global climate data, available at a resolution of 0.5°. These files include gridded information on rainfall, Tmx and Tmn. They are often utilized because of their comprehensive geographic and chronological scope (Harris et al. 2020; Hassan et al. 2020). The CRU database originated from data collected at 4,000 global weather stations. It was subjected to a two-stage quality control process involving initial consistency checks and subsequent removal of inaccuracies during interpolation (Shiru et al. 2020). The CRU monthly rainfall, Tmx and Tmn datasets were downloaded from https://crudata.uea.ac.uk/cru/data/hrg/

2.3 Global climate models

This study used simulations of four CMIP6 GCMs (O'Neill et al. 2016) for four shared socioeconomic Pathways (SSP126, SSP245, SSP370, and SSP585), as listed in Table 1. These four GCMs were selected as the preliminary assessment revealed their better performance in simulating Nigeria's climate. The CMIP6 GCMs data were downloaded from https://esgf-node.llnl.gov/search/cmip6/

3 Methods

The outputs of rainfall, Tmx, and Tmn from four GCMs were subjected to downscaling using the following methodologies.

1. The chosen GCMs underwent a regridding process to achieve a resolution of 0.5° × 0.5° using the bilinear interpolation approach to align the GCMs with the resolution of the reference data (CRU).
2. Thirteen distinct bias correction (BC) methods were applied to rectify the bias in the regridded GCMs for rainfall, Tmx, and Tmn throughout the period spanning from 1975 to 2014.
3. The initial assessment was conducted to reveal the effectiveness of various BC tactics. Subsequently, the six most promising procedures were identified using three statistical metrics: relative standard deviation (rSD), percentage of bias (PBIAS), and normalized root mean square error (NRMSE).

Table 1 The GCMs downscaled and bias corrected in the study

GCM	Developing Centre	Spatial resolution
ACCESS-ESM1-5	Australia Research Council Community	1.25° × 1.875°
BCC-CSM2-MR	Beijing Climate Centre China	1.125° × 1.125°
CMCC-ESM2	Euro-Mediterranean Centre Italy	0.942° × 1.25°
MRI-ESM2-0	Metrological Research Institute, Japan	1.125° × 1.125°

4. By employing multi-criteria decision analysis, the performance of the six BC approaches was assessed to determine the most favorable alternative.
5. The regridded GCM future simulation of rainfall, Tmx, and Tmn were corrected using the most optimal BC.
6. An MME was constructed using four downscaled GCMs to project rainfall, Tmx and Tmn for the near-term (2015–2059) and distant-term (2060–2099) periods in comparison to the historical period (1975–2014).

3.1 Bias correction methods

3.1.1 Power Transformation (PT)

This BC method uses a nonlinear approach in the form of exponential, e.g., logarithm, square root, etc., to make GCMs and observed data comparable (Leander & Buishand 2007; Wilcke et al. 2013). The monthly quantity of precipitation P_m or temperature T_m is converted to a corrected value of P* or T*in this approach using the equation below.

$$P^* = aP_m^b \tag{1}$$

$$T^* = aT_m^b \tag{2}$$

where b and a represent the exponent in the month and scaling factor, respectively, to ensure the adjusted precipitation or temperature matches the observed data, scaling factors "a" are computed based on the equation below,

$$a = \frac{\mu\left(P_{obs,m}\right)}{\mu\left(P^b\right)} \tag{3}$$

where μ represents the mean and $P_{obs,m}$ the observed monthly precipitation. When the value of b exceeds one then it suggests that the corrected temperature or precipitation underestimates its variability coefficient for month m.

3.1.2 Generalized Quantile Mapping (GQM)

GQM represents an expansion of quantile mapping, utilizing a more adaptable mapping function accounting for possible nonlinearity within the dataset (Cannon et al. 2015). it employs an overall Pareto distribution (GPD) to values over the threshold and a gamma distribution to values below the threshold (Heo et al. 2019; D.-I. Kim et al. 2019). GQM performed better in BC of extreme values and cases of non-linearities data (Ruffault et al. 2014). The GQM equation can be written as:

$$P_o = F^{-1}(F_m(P_m)) \tag{4}$$

where P_o is the bias corrected precipitation. F^{-1} is the inverse of the cumulative distribution function(CDF), F_m is the CDF of GCM raw precipitation, P_m. The gamma distribution and GPD are used in place of the pdf in this equation. GPD has an extreme value distribution (Shiru & Park 2020).

$$\Pr(X - u \le x | X > u) = \begin{cases} 1 - \left(1 + \frac{\xi x}{\tilde{\sigma}}\right), & if\ \xi \neq 0 \\ 1 - \exp\left(-\frac{\xi x}{\tilde{\sigma}}\right), & if\ \xi = 0 \end{cases} \tag{5}$$

In this approach, a threshold denoted by u is defined by the 95th percentile value. The scale parameter, represented as $\tilde{\sigma}$, is calculated as $\tilde{\sigma} = \sigma + (u - \mu)$, and ξ is the shape parameter. The gamma distribution is used this way on values below a lesser threshold (below the 95th percentile), and the GPD is used on values exceeding this threshold.

$$\begin{cases} F^{-1}_{obs,\ gamma}\left(F_{m,\ gamma}\right) & if\ x\ < 95th\ percentile \\ F^{-1}_{obs,\ GPD}\left(F_{m,\ GPD}\right) \end{cases} \tag{6}$$

3.1.3 Scaling method (SCLM)

This method objectively corrected the long-term monthly GCM values to align with those of the observed data (Daniel 2023; Islam et al. 2023). This method can effectively account for climatic conditions when incorporating monthly mean values between the GCMs and observed data (Nashwan et al. 2020). Typically, multiplication factors and additional terms are employed in the correction of rainfall and temperature, respectively, and these corrections are applied every month (Lafon et al. 2013). The equation can correct the monthly precipitation:

$$P_{cor,m} = P_{GCMs,m} \times \frac{\mu\left(P_{obs,m}\right)}{\mu\left(P_{GCMs}\right)} \tag{7}$$

The Temperature, T is corrected using the following equation,

$$T_{cor,m} = T_{GCMs,m} + \mu\left(T_{obs,m}\right) - \mu\left(T_{GCMs,m}\right) \tag{8}$$

where $P_{cor,m}$ and $T_{cor,m}$ are the corrected rainfall and temperature of the mth month, $\mu\left(P_{obs,m}\right)$ and $\mu\left(T_{obs,m}\right)$ denotes the average value of observed rainfall and temperature. $T_{GCMs,m}$ and P_{GCMs} depicts the GCMs of rainfall and temperature of mth month.

3.1.4 Empirical Quantile Mapping (EQM)

Empirical quantile mapping is a BC technique in which a model's cumulative distribution function (CDF) is fitted to that of observation data (Enayati et al. 2021; X. Li & Babovic 2019). Its fundamental concept is based on constructing empirical cumulative distribution functions (ECDFs) daily point by point. EQM is useful for correcting biases in simulated models because it primarily aims to maintain the observed distribution (Amengual et al. 2012; Wilcke et al. 2013). The equations can correct the monthly temperature and rainfall:

$$T_{cor,m} = ecdF^{-1}_{obs,m}\left(ecdF_{hst,m}\left(T_{hst,m}\right)\right) \tag{9}$$

$$P_{cor,m} = ecdF^{-1}_{obs,m}\left(ecdF_{hst,m}\left(P_{hst,m}\right)\right) \tag{10}$$

where $ecdF^{-1}_{obs}$ is the empirical inverse CDF of observed data and $ecdF_{hst}$ is the empirical CDF of GCM historical simulation.

3.1.5 Delta change method (DM)

The Delta change method is one of the simple BC techniques (Sarr et al. 2015). It is based on the total anomalies interpolated to resolution on monthly climate Surfaces (Mendez et al. 2020). Delta change generates climatic scenarios by adding the future change from the GCMs model for the observation data's perturbations (Teutschbein & Seibert 2012). The disparity, i.e., delta between the historical observations and GCMs future projections on a coarse scale, is calculated, and the delta obtained is added to historical observation (Luo et al. 2018). it assumes a constant change in climate variables between historical and future periods (Beyer et al. 2020).

$$P^{BC}_{contr}(t) = P_{obs}(t) \tag{11}$$

$$P^{BC}_{frc}(t) = P_{obs}(t).\left[\frac{\mu_m\left(P_{frc}(t)\right)}{\mu_m\left(P_{contr}(t)\right)}\right] \tag{12}$$

$$T_{sim}(t)' = T_{obs}(0) + \left(T_{sim}(t) - T_{sim}(0)\right) \tag{13}$$

where P is the monthly rainfall and "contr" is the simulated GCMs, "obs" represents the observation, "frc" represents the future GCMs, and "BC" represents bias corrected GCM. t is the time step, μ_m is the long-term yearly mean. $T_{sim}(t)'$ represents the simulated temperature based on present observation, $T_{sim}(t)$ is the simulated monthly temperature.

3.1.6 Quantile Delta Mapping (QDM)

QDM is a statistical BC technique that focuses on the changes (deltas) in quantiles between historical observed data and the future projections of climate models (Casanueva et al. 2020; Gori et al. 2022). To generate future projections, the changes are added to the observed historical data (Cannon et al. 2015). This technique involves (i) detrending from individual quantiles, (ii) applying quantile mapping (QM) to detrended data, and (iii) reintroducing trends to bias-adjusted quantiles, preserving change signals in all quantiles. This enables both multiplicative and additive corrections for rainfall and temperature, respectively. The results in BC for the future period at time t equals this BC $\hat{x}$ multiplied with the precipitation change $\Delta(t)$ and added for absolute change in $\Delta(t)$ temperature. It is given as,

$$x_{bc}(t) = \hat{x}(t)\Delta(t) \tag{14}$$

$$x_{bc}(t) = \hat{x}(t) + \Delta(t) \tag{15}$$

For rainfall, the relative change $\Delta(t)$ is given below:

$$\Delta(t) = \frac{x_{m,p}(t)}{F^{-1}_{m,c}\left\{F^{(t)}_{m,p}\left[x_{m,p}(t)\right]\right\}} \tag{16}$$

Also for temperature, the relative change $\Delta(t)$ can be written as

$$\Delta(t) = x_{m,p}(t) - F^{-1}_{m,c}\left\{F^{(t)}_{m,p}\left[x_{m,p}(t)\right]\right\} \tag{17}$$

where $x_{m,p}$ is the monthly raw GCM temperature series, and $x_{bc}(t)$ is the bias-corrected series.

3.1.7 Variance Scaling (VarS)

VarS corrects biases in climate models and improves local-scale projections by adjusting the variability of climate model data to match that of observed historical data (Maraun 2016). Biases in climate models that may underrepresent climatic variability are addressed via variance scaling (Maraun et al. 2010). Adjusting the variance of the model data while maintaining the mean value (Piani et al. 2010) reduces these biases and enhances the models' capacity to represent observed climatic variability (Teutschbein & Seibert 2012). Variance scaling presumes data stationarity (Koutsoyiannis 2006), but due to climate change, this may not be the case for non-stationary variables (Dixon et al. 2016). It focuses mostly on variability while posing potential constraints on extreme events. The selection of scaling factors and data quality are essential.

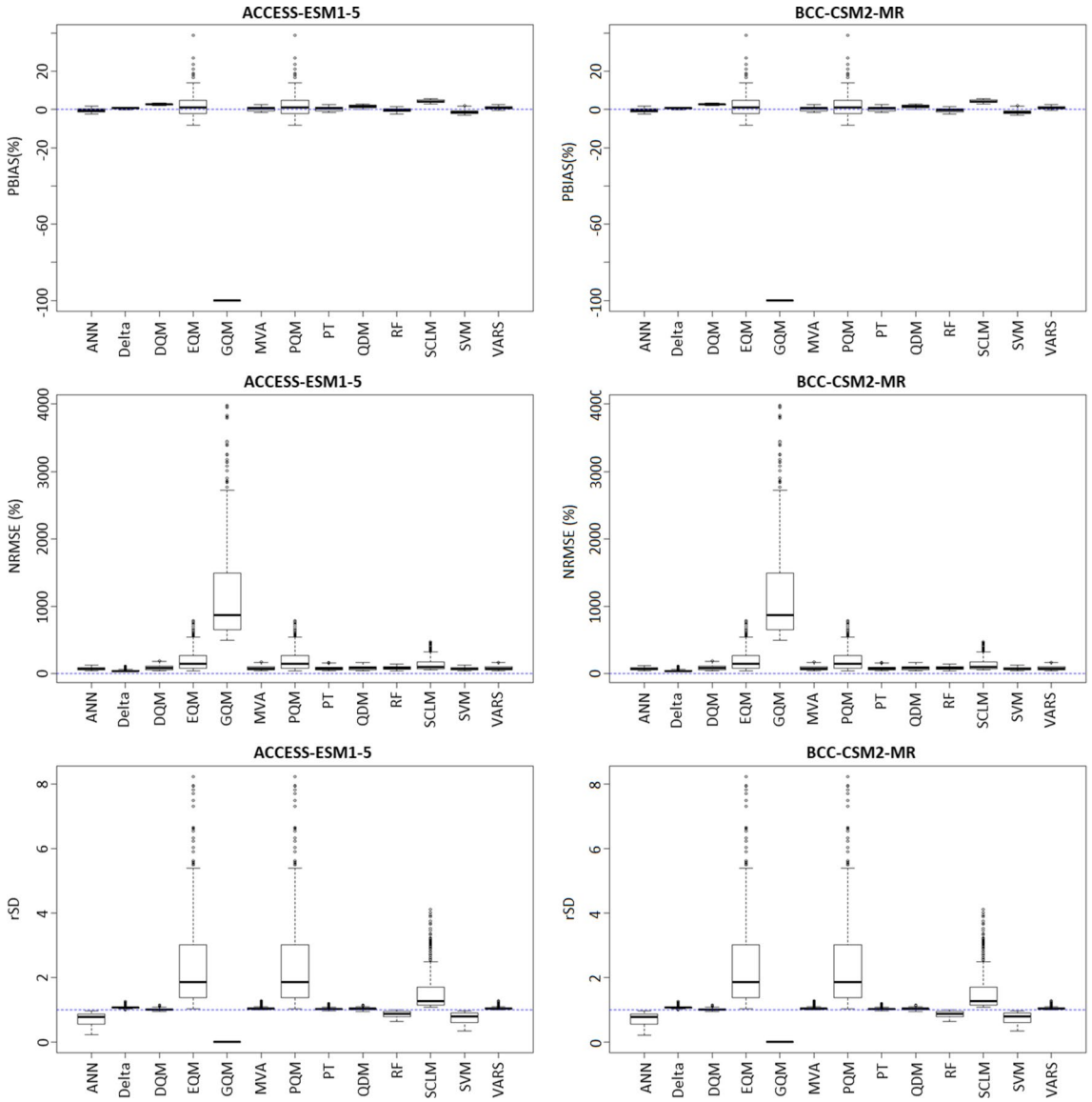


Fig. 2 The preliminary assessment of 13 techniques in bias correcting ACCESS-ESM1-5 and BCC-CSM2-MR rainfall

$$T_{cor,m} = \left[T_{raw,m} - \mu\left(T_{raw,m}\right)\right] \times \frac{\sigma\left(T_{obs,m}\right)}{\sigma\left(T_{raw,m}\right)} + \mu\left(T_{obs,m}\right) \quad (18)$$

where $T_{cor,m}$ corrected monthly temperature series, $T_{raw,m}$ is the GCM raw temperature series, and $T_{obs,m}$ is the observed series.

3.1.8 Parametric Quantile Mapping (PQM)

PQM involves modeling the climate model output's distribution using parametric distributions (such as the Gaussian distribution) and subsequently mapping the quantiles of the model's distribution with those of the observed historical data (Passow & Donner 2020). PQM allows for more accurate BC because it presumes that the model data can be characterized by a particular probability distribution (Switanek et al. 2017). When model data closely matches a known distribution, it can effectively correct bias. Still, if model data deviates significantly from the selected distribution, it may be less effective.

3.1.9 Mean and Variance Adjustment (MVA)

MVA is a simplified statistical method that adjusts the variance and the mean of climatic model data to align with those of reference observed historical data (Gutjahr & Heinemann 2013; Wilcke et al. 2013). It harmonizes the GCM data to enhance their conformity with

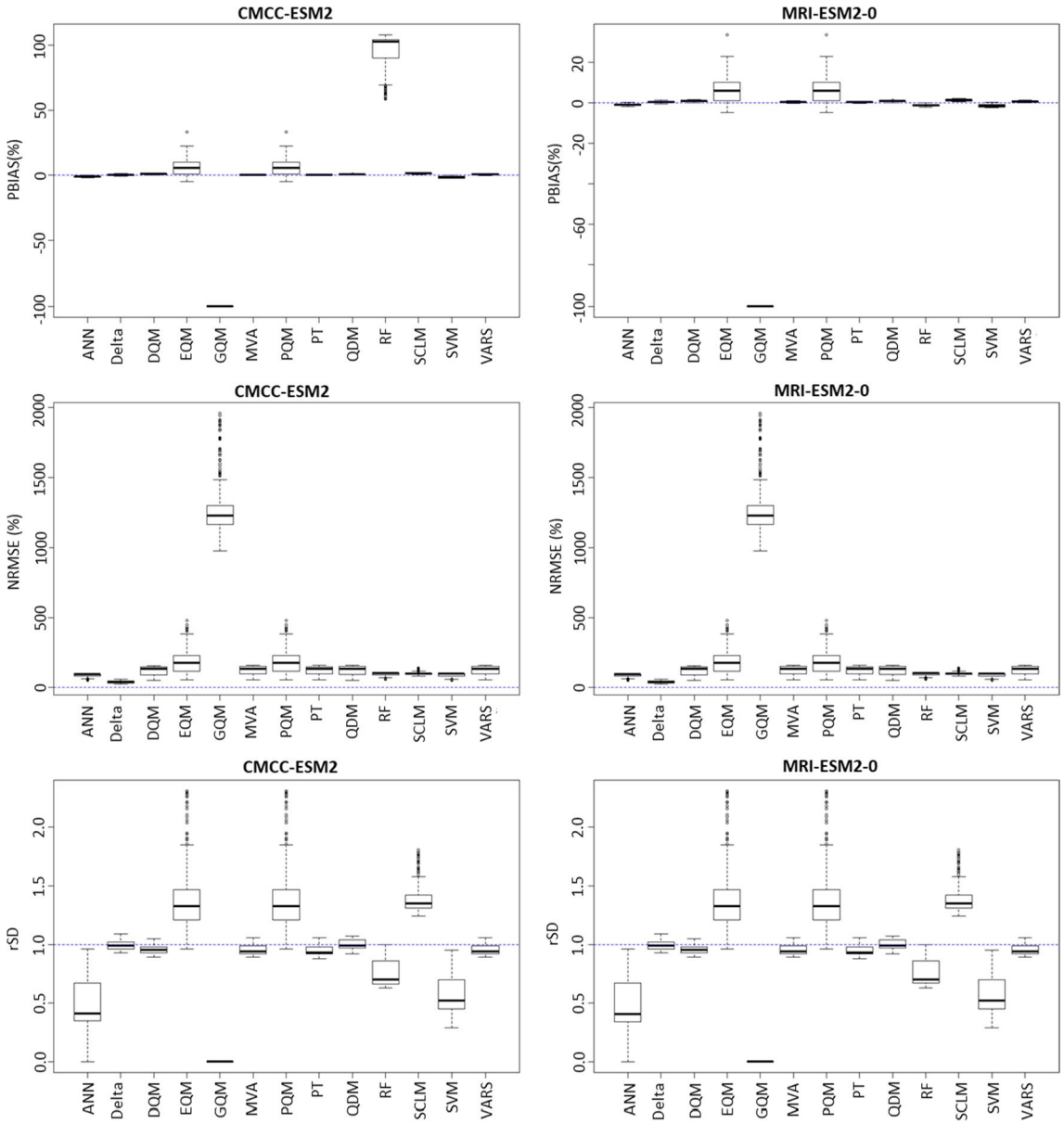


Fig. 3 The preliminary assessment of 13 techniques in bias correcting CMCC-ESM2 and MRI-ESM2-0 rainfall

the statistical characteristics of observed data (Luo et al. 2018). While MVA offers simplicity and ease of implementation, it may not address all bias aspects, mainly when dealing with non-normally distributed data (Wilcke et al. 2013).

3.1.10 Detrended Quantile Matching (DQM)

DQM, as a statistical BC method, reduces seasonal variations and longer-term patterns from climate data (Maraun et al. 2019). It then generates bias corrected projections by comparing large-scale climatic variables quantiles (percentiles) to the quantiles of observed historical data (Ali et al. 2019). It accounts for trends and biases in climate projection without altering the distributional properties of observed data (Wilcke et al. 2013).

3.1.11 Artificial Neutral Network (ANN)

ANN establishes computational models based on the functioning of the human brain, which is composed of several nodes and interconnected layers of processing units (neurons) (Okwu et al. 2021). ANNs are regression-based statistical techniques used to establish complex, nonlinear relationships between GCM simulated variables (predictors) and

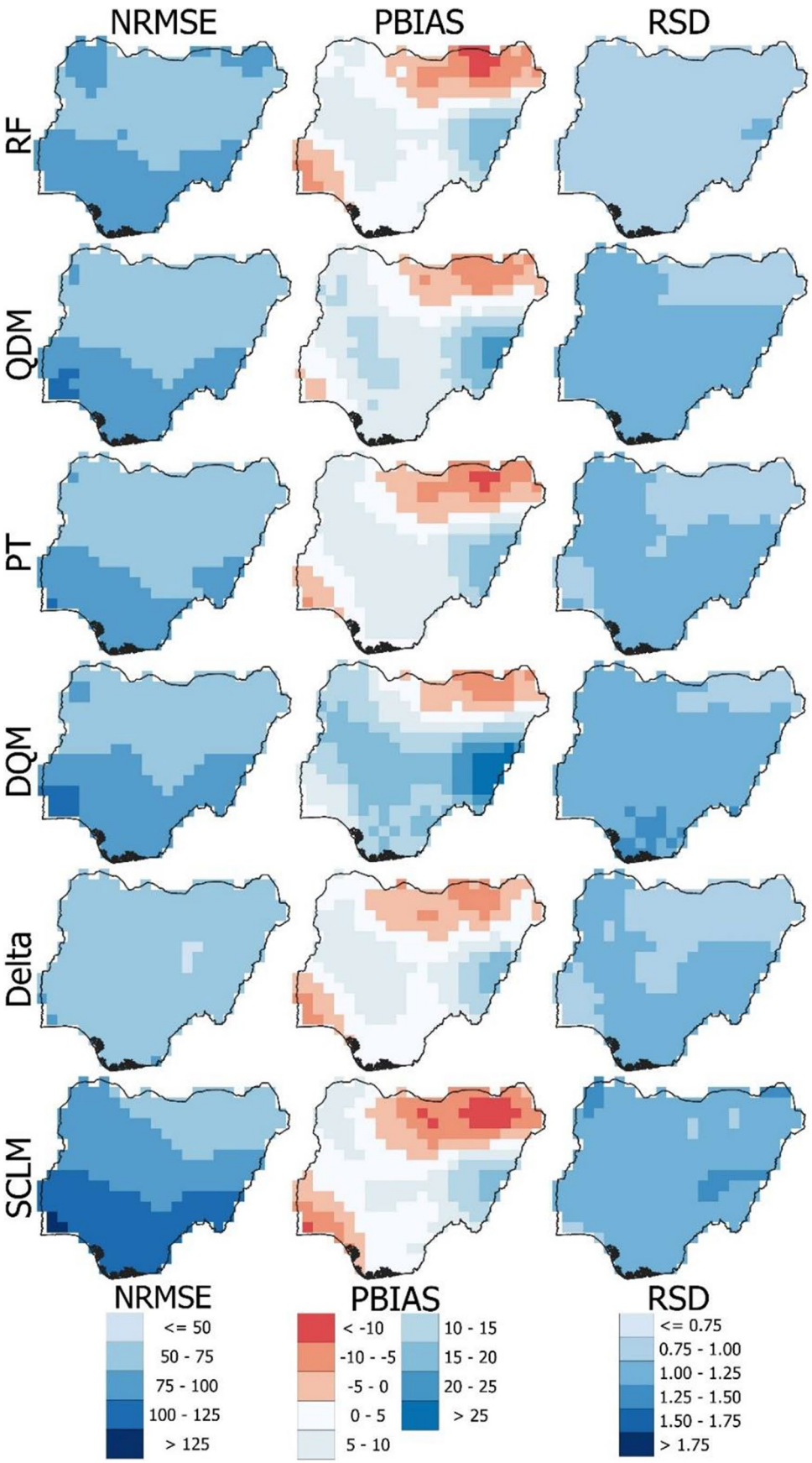


Fig. 4 Spatial distribution of the performance of bias correction models based on three statistical metrics in downscaling ACCESS-CM2 rainfall

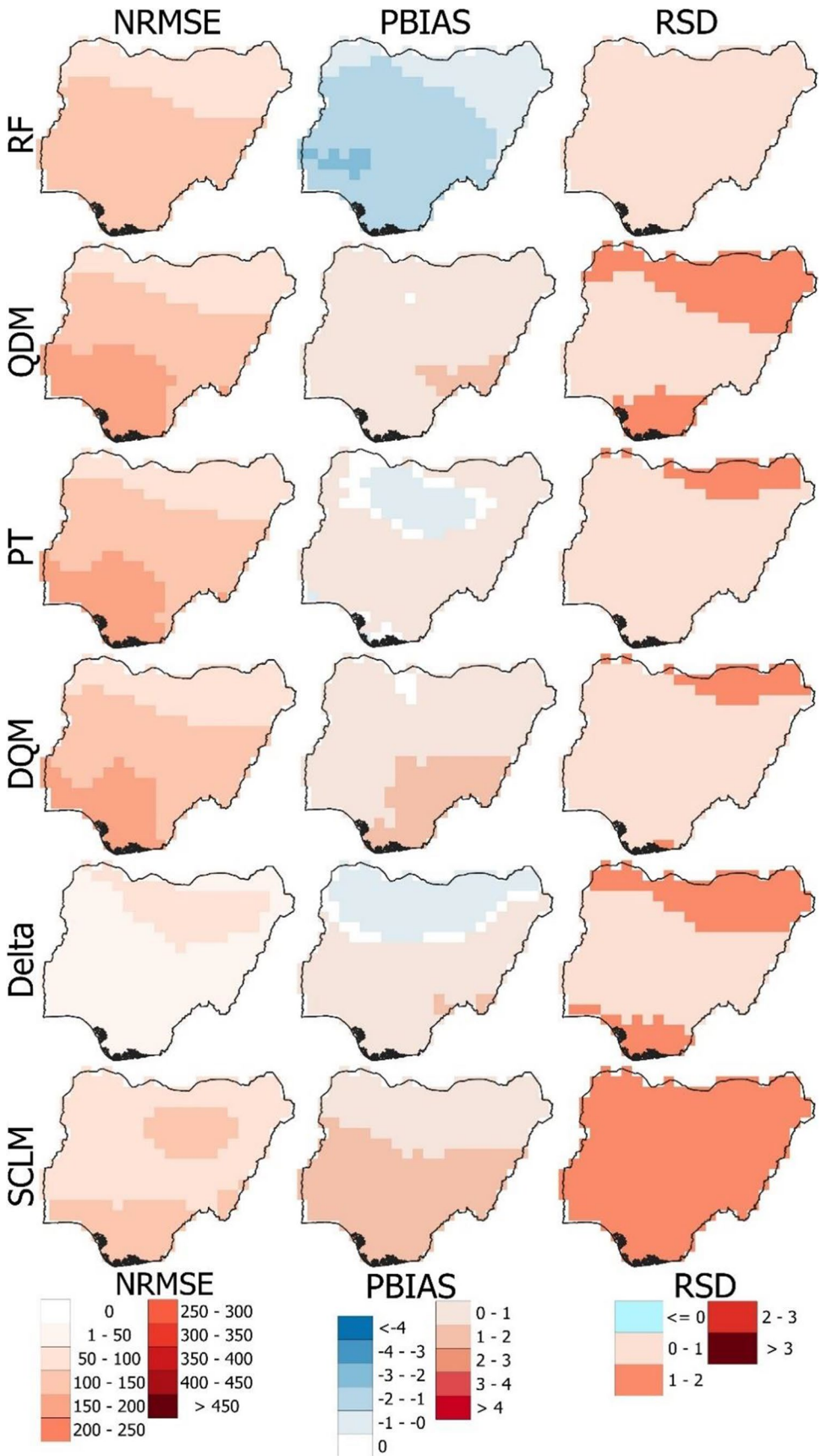


Fig. 5 Spatial distribution of the performance of bias correction models based on four statistical metrics in downscaling ACCESS-CM2 maximum temperature

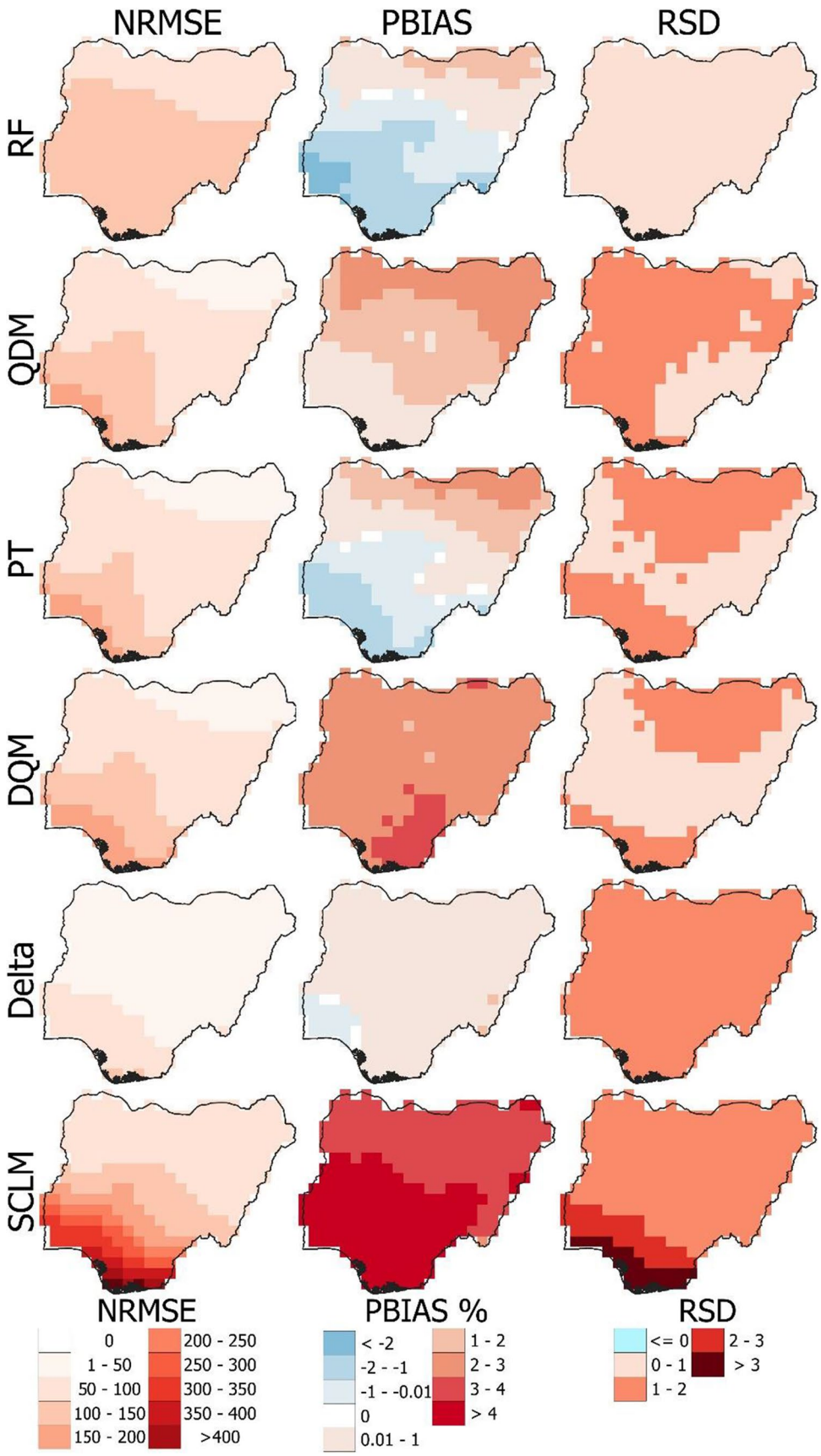


Fig. 6 Spatial distribution of the performance of bias correction models based on four statistical metrics in downscaling ACCESS-CM2 minimum temperature

Table 2 Compromise programming index of different downscaling methods based on their performance in terms of three statistical metrics in downscaling rainfall of four selected GCMs

GCM	NRMSE				PBIAS				RSD				CPI
Method	ACCESS-ESM1-5	BCC-CSM2-MR	CMCC-ESM2	MRI-ESM2-0	ACCESS-ESM1-5	BCC-CSM2-MR	CMCC-ESM2	MRI-ESM2-0	ACCESS-ESM1-5	BCC-CSM2-MR	CMCC-ESM2	MRI-ESM2-0	
DM	41.5	41.4	41.5	41.5	3.3	-1.1	2.0	2.7	1.0	1.0	1.0	1.0	0.0
DQM	60.0	61.4	55.4	74.0	11.5	2.4	-0.1	1.9	1.1	1.0	1.1	1.0	0.8
PT	55.4	64.2	54.1	70.9	4.1	0.1	-1.1	2.4	1.0	0.9	1.0	1.0	0.7
QDM	55.9	60.7	54.7	73.2	6.3	3.2	-0.7	2.6	1.1	1.0	1.0	1.0	0.8
RF	61.4	64.9	52.0	66.5	3.2	2.7	-2.7	-1.2	0.9	0.9	0.9	0.9	0.6
SCLM	61.7	98.4	55.0	60.6	1.3	-9.0	-1.7	-0.3	1.1	1.2	1.1	1.1	1.0

local climate variables (predictands) (Nourani et al. 2019). They are trained based on historical data and reduce the biases to adjust the error to a minimal value (Gutjahr & Heinemann 2013). Prediction of local climate variables from large-scale climate inputs can be performed once the ANN has been trained (Piani et al. 2010). ANNs exhibit remarkable flexibility and versatility, accommodating a wide range of data types, and also excel at capturing complex data relationships and adjusting to intricate patterns. Training ANNs demand significant computational resources and substantial training data (Wilcke et al. 2013).

3.1.12 Random Forest (RF)

RF is a categorization method applied in various research on the assessment of climate change.

RF combines the output of multiple decision trees, generated from a bootstrap sample, to arrive at a single result (Raczko & Zagajewski 2017). RF is less prone to overfitting than a decision tree due to bootstrap sampling, even if the decision tree approaches are created by partitioning the training data (Wilcke et al. 2013). Multiple decision trees are assembled in this ensemble learning technique to capture intricate interactions between predictor and predictor parameters (Gutjahr & Heinemann 2013). RF can effectively manage complex and nonlinear data relationships, demonstrating strong performance with extensive and multidimensional datasets.

3.1.13 Support Vector Machine (SVM)

SVM is a supervised learning method that aims to identify a function that optimally matches the training data while maximizing the error tolerance margin (Cortes and Vapnik 1995). This results in resilient predictions that are less influenced by outliers. SVM create hyperplanes that effectively separate data points by maximizing the margin, providing flexibility in accurately fitting the data. In the case of nonlinear data, kernels are employed to convert it into a higher-dimensional space, enabling linear separation. The support vectors, which are the crucial data points defining the hyperplanes, directly impact the model's prediction function. SVM-based regression models prioritize the maximization of the margin, resulting in reduced sensitivity to outliers compared to alternative regression approaches. It has strong performance in processing data with a high number of dimensions, even when the number of training samples is restricted. SVM utilizes kernels to represent intricate, nonlinear correlations within the dataset effectively. Consequently, it has been determined to be particularly effective in resolving nonlinear issues in climate downscaling (Pour et al. 2018).

Table 3 Compromise programming index of different downscaling methods based on their performance in terms of three statistical metrics in downscaling rainfall, maximum temperature and minimum temperature of four selected GCMs

Method	Rainfall		Tmx		Tmn		Average Rank	Overall Rank
	CPI	Rank	CPI	Rank	CPI	Rank		
DM	0.00	1	0.44	3	0.35	5	3.0	3
DQM	0.77	5	1.00	6	0.31	3	4.7	5
PT	0.74	3	0.36	2	0.29	2	2.3	2
QDM	0.76	4	0.72	5	0.34	4	4.3	4
RF	0.62	2	0.00	1	0.00	1	1.3	1
SCLM	1.00	6	0.50	4	1.00	6	5.3	6

The conventional bias correction methods, PT, GQM, Scaling, EQM, DM, QDM, VarS, PQM, MVA, and DQM, were implemented using the 'downscaleR' package. The 'nnet', 'e1071', and 'randomForest' packages of the statistical software R were employed to implement ANN, SVM, and RF bias correction methods, respectively.

3.2 Compromise programming for ranking the bias correction techniques

This study explores compromise programming, an approach to decision-making based on the Euclidean distance metric (Raju & Kumar 2014). The statistical metrics above were employed to select the best BC technique using the compromise method. The CP method determines the most suitable BC approach based on the smallest distance between the bias-corrected and an ideal value within the available dataset (Salman et al. 2019). The CP calculates the distance Lp metric using the values of the statistical metrics.

$$L_p = \left[\sum_{i=1}^{n} \left|f_j^* - f_j\right|^p\right]^{1/p} \tag{19}$$

where L_p represents the CPI, f_j^* is the ideal performance value, and f_j is the actual performance measure. The parameter p can have any value in the range from 1 to ∞. Nonetheless, values 1, 2, and ∞ are the most frequently employed (Muhammad et al. 2019). The lower the value of CPI, the better the performance of the bias correction method.

The CPI of BC techniques were computed from the three statistical metrics, NRMSE, PBIAS and rSD, in correcting rainfall, Tmx and Tmn of four GCMs. The mathematical equations of these metrics are:

$$NRMSE = \frac{\left[\frac{1}{n}\sum_{i=1}^{n}\left(x_{sim,i} - x_{obs,i}\right)^2\right]^{1/2}}{\frac{1}{n\sum_{i=1}^{n}x_{sim,i}}} \tag{20}$$

$$\text{Pbais} = \left[\frac{\sum_{i=1}^{n}\left(x_{obs} - x_{sim}\right)}{\sum_{i=1}^{n}x_{obs}}\right] \times 100 \tag{21}$$

$$\text{rSD} = \frac{SD\left(x_{sim}\right)}{SD\left(x_{obs}\right)} \tag{22}$$

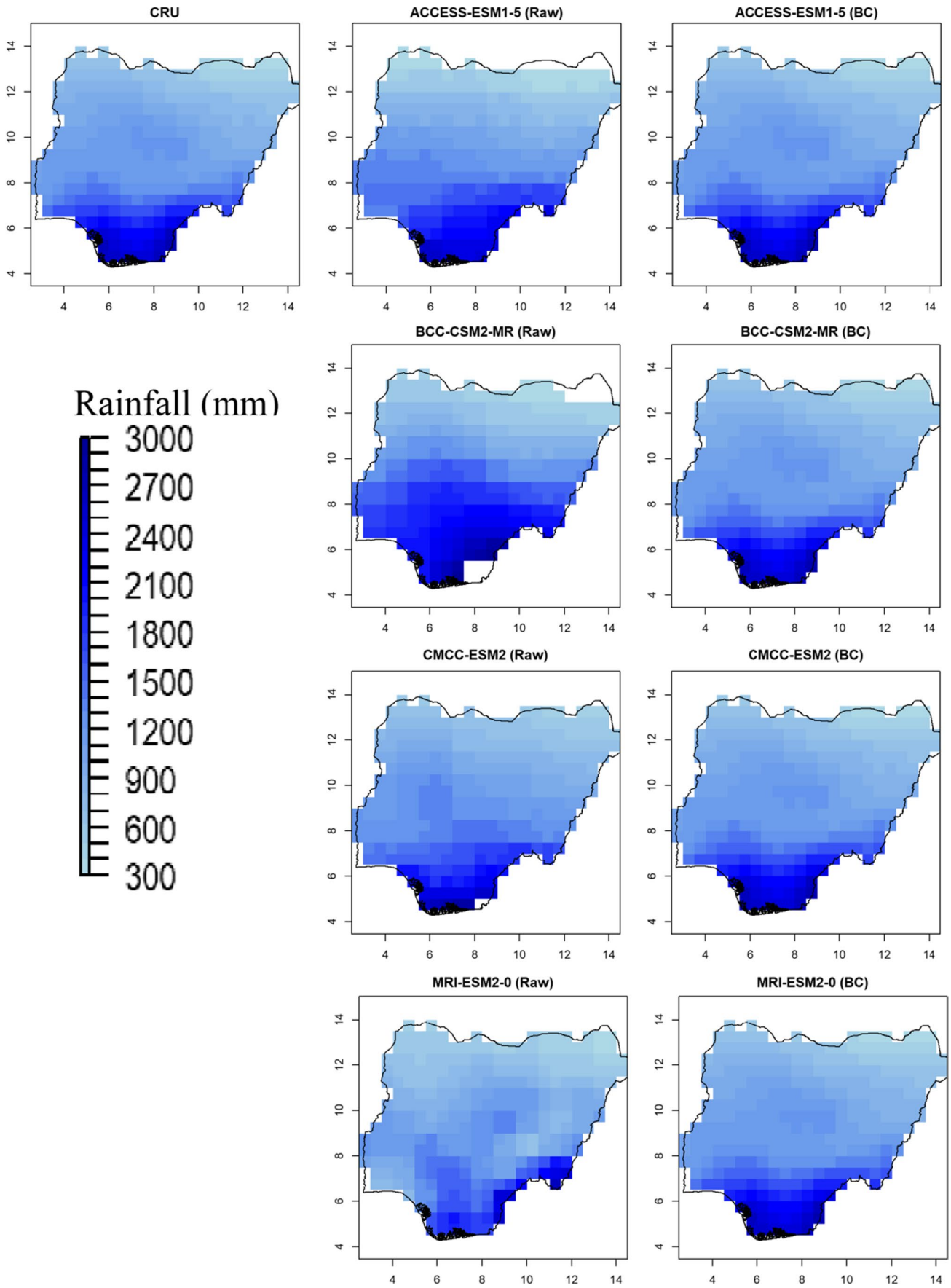
where $x_{sim,i}$ and $x_{obs,i}$ are the simulated and observed values, respectively. σ and μ are the standard deviation and mean of the observation values. The 'hydroGOF' package was used to assess the statistical performance of these bias correction methods.

The metrics were standardized based on their ideal values to reduce the influence of the high-value range. The ideal values for NRMSE, PBIAS, and rSD are 0, 0, and 1, respectively. So, all the values were standardized between 0 and 1. For NRMSE and PBIAS, the lower value was selected and subtracted from all the values, while for rSD, the higher value was selected and subtracted from all the values. The square values obtained were finally standardized using the formula,

$$CPI = \frac{Value - \min\ value}{\max\ value - \min\ value} \tag{22}$$

3.3 Multimodel Ensemble (MME)

MME integrates the outputs of multiple GCMs to generate a more complete and dependable future climate projection (Ahmed et al. 2020). Every GCM contains inherent uncertainties arising from simplifications, assumptions, and parameterizations. MMEs mitigate the effects of these uncertainties and yield a more resilient estimation of the probable spectrum of future climate projections. This study created the MME by computing the average of the outputs from four selected GCMs. The MME is thereafter employed to project precipitation and temperature in Nigeria for various SSPs.


CRU
ACCESS-ESM1-5 (Raw)
ACCESS-ESM1-5 (BC)
BCC-CSM2-MR (Raw)
BCC-CSM2-MR (BC)
CMCC-ESM2 (Raw)
CMCC-ESM2 (BC)
MRI-ESM2-0 (Raw)
MRI-ESM2-0 (BC)
Rainfall (mm)
3000
2700
2400
2100
1800
1500
1200
900
600
300

◂**Fig. 7** Spatial distribution of observed, raw GCM and bias-corrected GCMs rainfall

4 Results

4.1 Preliminary assessment of bias-correction techniques

Figures 2 and 3 show a preliminary assessment of several BC strategies using box plots. The evaluation involved using the three statistical measures to compare the BC techniques results from the four GCM historical simulations and CRU data (reference dataset) at each grid point. In the boxplot, an ideal line was utilized to select BC strategies; the line was drawn for the three metrics depending on their ideal value. The optimal Pbias, NRMSE, and rSD values were 0 and 1, respectively. Each box depicts the statistical measure obtained for a GCM using a specific BC technique. Most BC strategies fared well in correcting the GCMs, while a few others performed poorly. The BC procedures closest to the ideal values were chosen for the four GCMs and three variables (rainfall, Tmx and Tmn). Six BC strategies were chosen based on the assessment results: the delta method, DQM, PT, QDM, RF, and SCLM.

4.2 Spatial distribution of the performance metrics

Figure 4 depicts the spatial distribution of the performance metrics in correcting the bias of ACCESS-CM2 rainfall data. According to the NRMSE, all BC strategies fared better in the country's northern region than in the southern part. While SCLM fared better than other strategies, the Delta method performed better.

All BC algorithms demonstrated varied degrees of improvement in the raw GCM in PBIAS. They also performed fairly well (under-rating) at the grid points in the Northeast region. The DQM approach fared the worst, while RF and Delta did the best. Regarding rSD measures, RF approaches beat other techniques in general improvement of the raw GCM, whereas SCLM did the worst.

Figure 5 displays the regional distribution of the three metrics on the highest temperature of the ACCESS-CM2. The NRMSE demonstrates that the Delta approach is the best, while QDM is the worst, and that the strategies better corrected the country's northern than the southern region. Regarding the Pbias methodology, the PT approach fared the best compared to the other ways, while the RF technique performed the worst. Regarding rSD, RF managed best because most of the grid points in the GCM were near those in the reference dataset, and SCLM did the worst.

Figure 6 shows the spatial distribution of the NRMSE in bias-correcting ACCESS-CM2 Tmn. All BC strategies performed better in the northern part compared to the southern region. The Delta approach outperformed the other techniques, whereas SCLM did the worst. Regarding Pbias metrics, the DQM technique is the best, while the SCLM technique is the worst, producing the most considerable inaccuracy in the country when bias correcting the GCM. Regarding rSD measures, RF approaches beat other methods in general improvement of the raw GCM, whereas SCLM and Delta are the least improved.

4.3 Ranking of the bias correction method

Each BC method was evaluated using three measures based on its performance in downscaling all four GCMs. Each BC technique yielded twelve measurements (three metrics × four GCMs). CPI was calculated for each BC approach using these twelve metrics. CPI was calculated for each BC approach using these twelve metrics. The CPI of different downscaling methods based on their performance in terms of three statistical metrics in correcting rainfall for four selected GCMs is presented in Table 2. The estimated CPI of different BC methods for rainfall is presented in the last column of the table. The table shows DM as the best method, followed by RF, PT, QDM, and DQM. The worst performance was using SCLM.

The same procedure was followed to estimate the CPI of different methods for correcting Tmx and Tmn. The obtained results for all three variables are presented in Table 3. The table shows the various performances of different BC methods for downscaling rainfall, Tmx, and Tmn. For example, DM shows the best performance in correcting rainfall, while RF shows the best in correcting both Tmx and Tmn. Therefore, the CPIs of the BC methods for the three variables were averaged to have a consensus ranking of the methods. The average CPIs of the methods are presented in the fifth column of the table. The average CPI ranges from 0.21 to 0.83 for different methods. RF showed the lowest CPI, while SCLM showed the highest. The ranking of the BC methods based on the average CPI is presented in the last column of the table. It shows the best performance of RF, followed by DM, PT, QDM, and DQM. SCLM showed the worst performance.

4.4 Evaluation of the random forest bias correction method

4.4.1 Spatial distribution pattern

The bias-corrected GCMs for rainfall, Tmx and Tmn were compared to the raw GCMs and CRU data using the

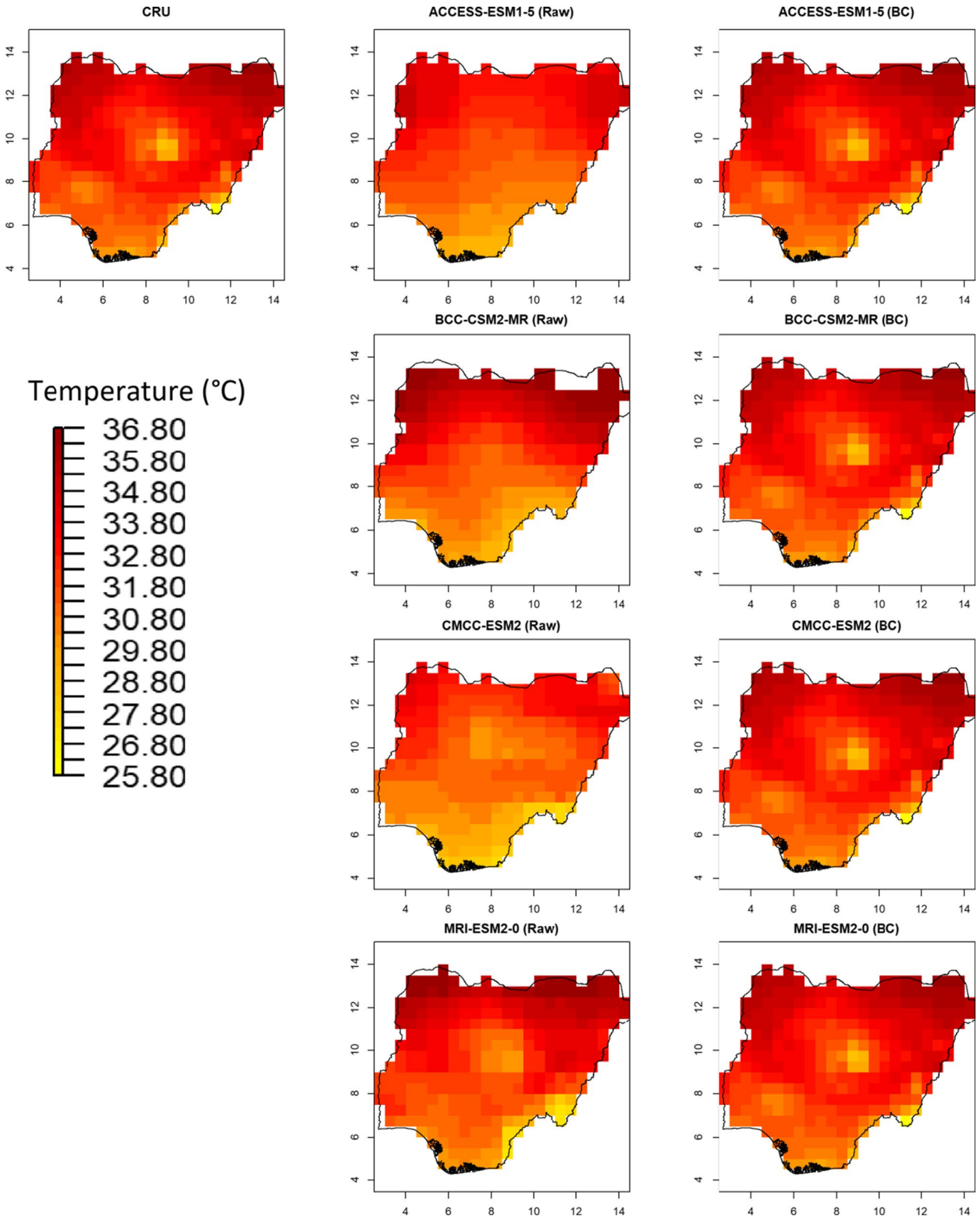


Fig. 8 Spatial distribution of observed, raw GCM and bias-corrected GCMs maximum temperature

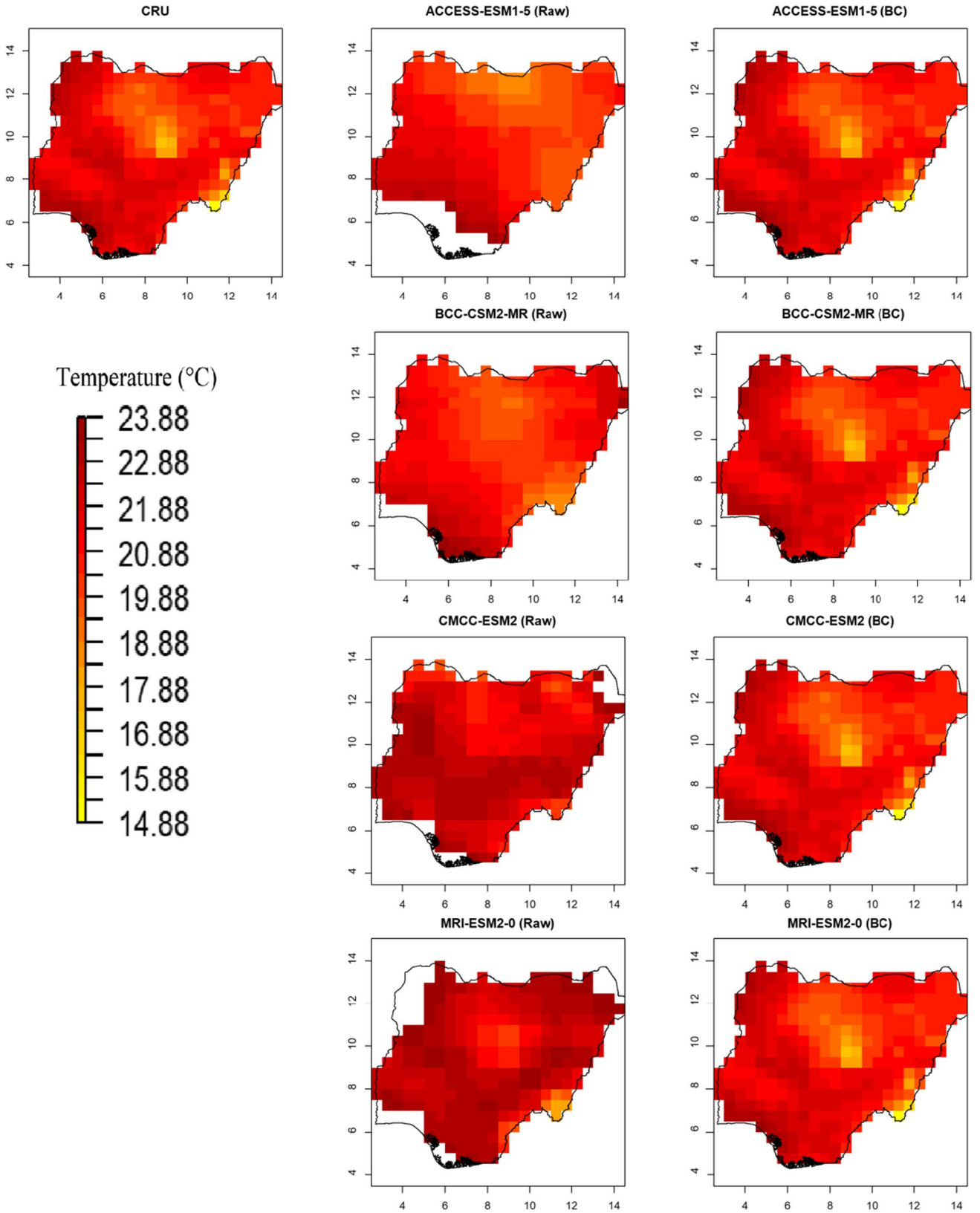


Fig. 9 Spatial distribution of observed, raw GCM and bias-corrected GCMs minimum temperature

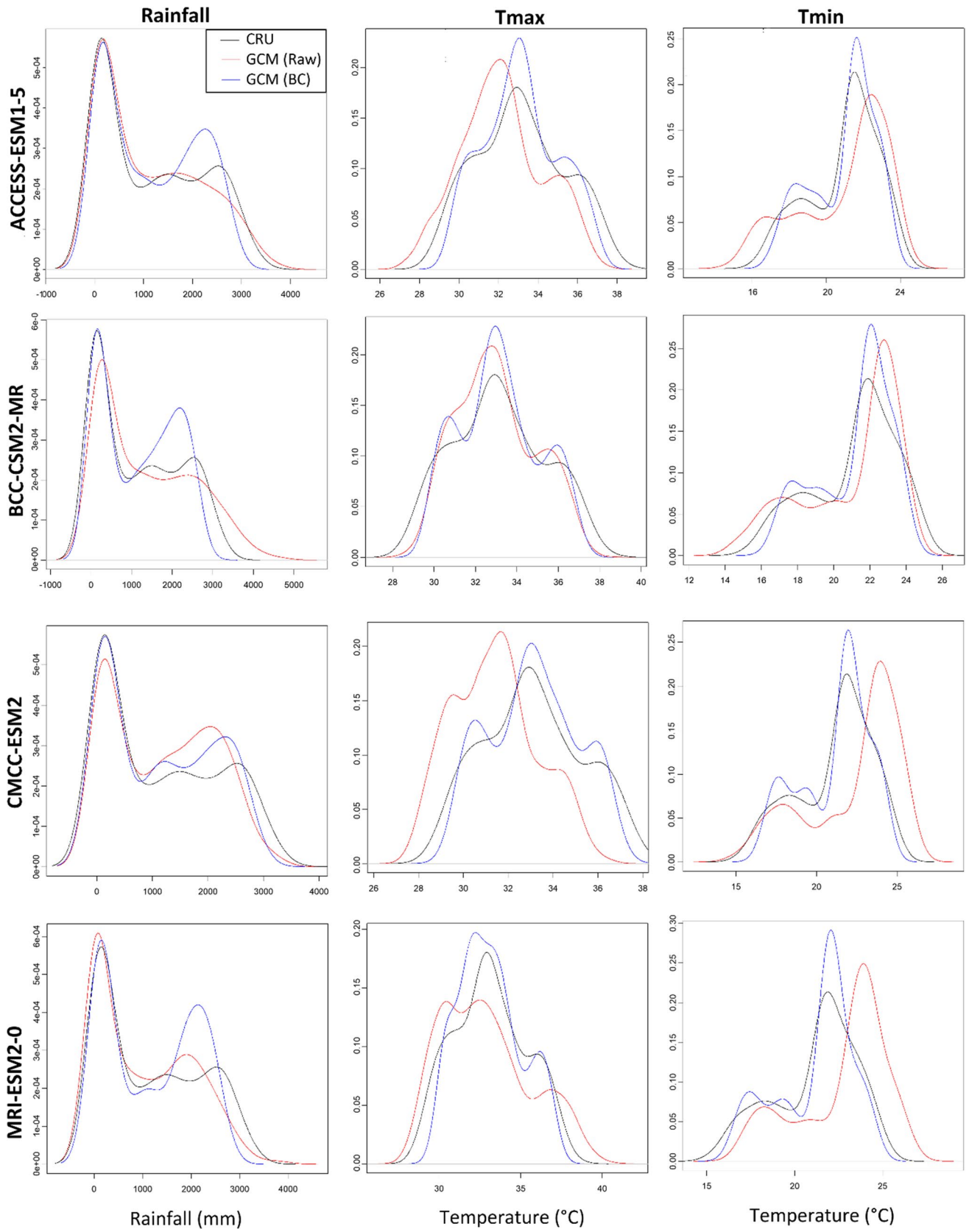


Fig. 10 Probability distribution function (PDF) of the monthly observed data, raw GCMs, and bias-corrected GCMs of rainfall, maximum temperature and minimum temperature in columns, while the rows represent the GCMs

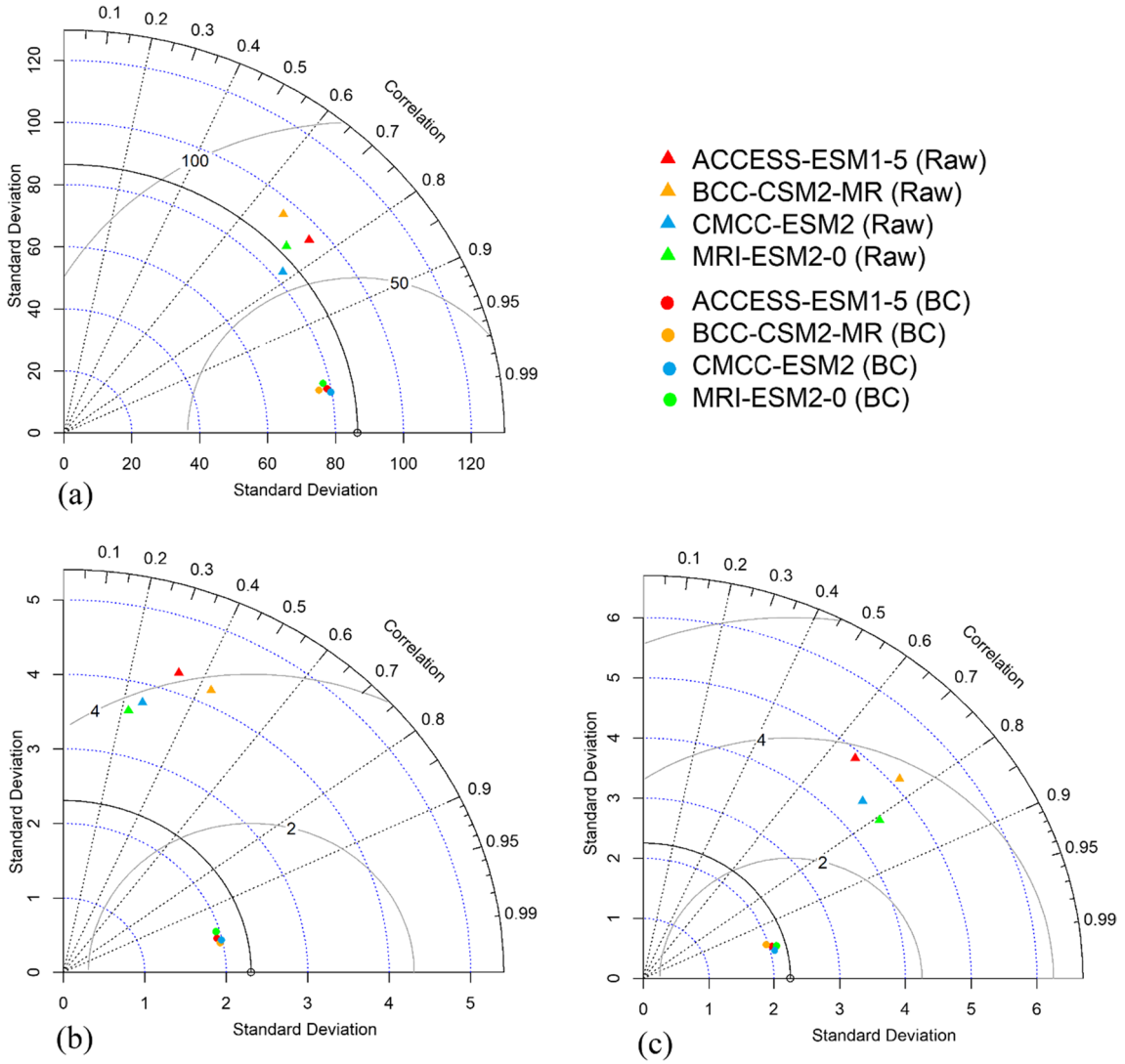


Fig. 11 Taylor diagram raw and bias-corrected GCMs of (**a**) rainfall, (**b**) maximum temperature, and (**c**) minimum temperature

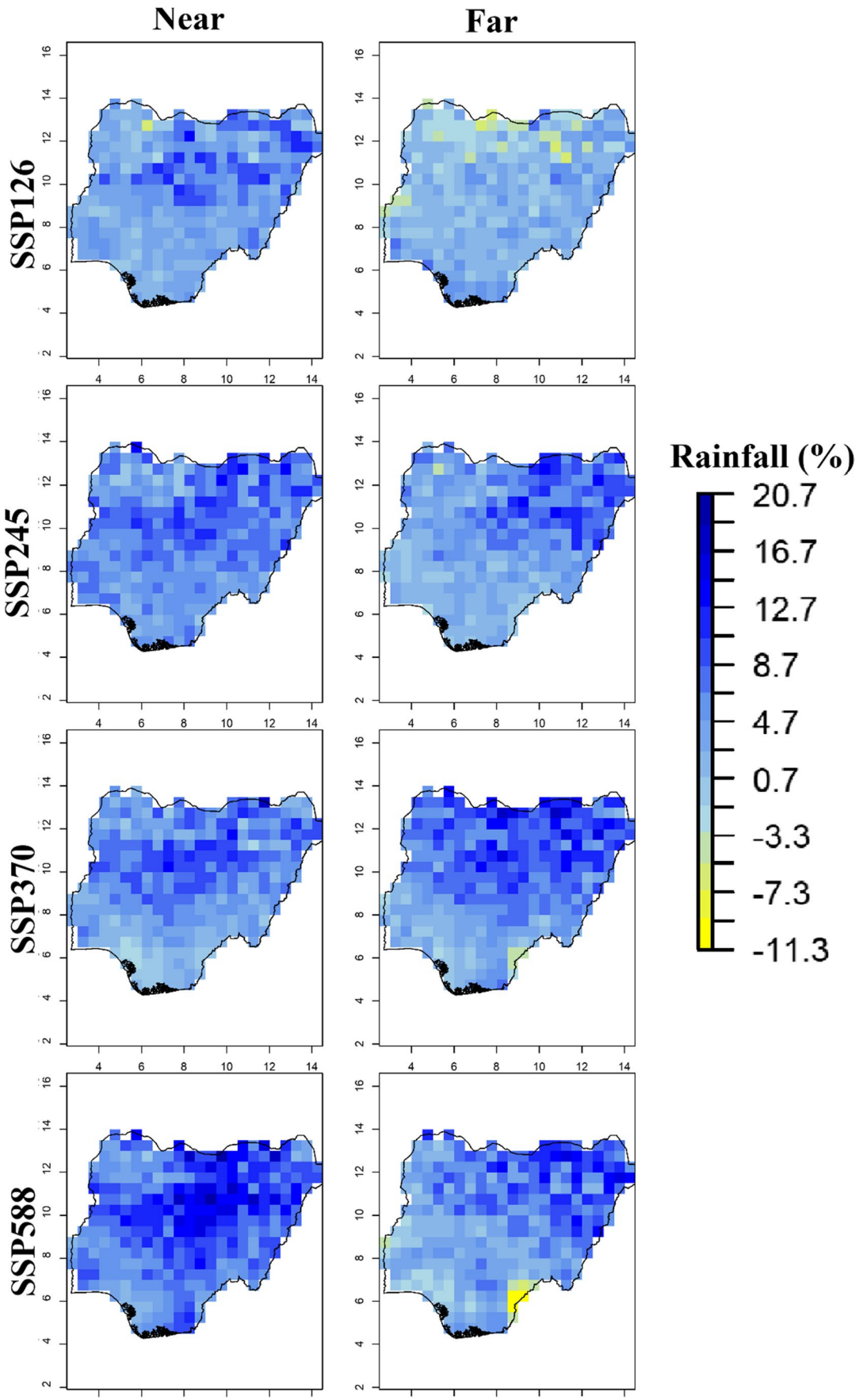
spatial distribution pattern depicted in Figs. 7, 8, and 9. The results demonstrate a high resemblance between the bias-corrected models and the CRU, indicating that the RF method enhanced the GCM' simulation and accurately depicted CRU spatial variability.

4.4.2 Probability density function of the bias-corrected GCMs

The PDF of the monthly raw, bias-corrected GCMs using random forest (RF) and historical (CRU) data is shown in Fig. 10. The blue and red lines reflect the bias-adjusted and raw GCM PDFs, respectively, while the black line represents the CRU PDF. Monthly rainfall, Tmx and Tmn data from all CRU grid points from 1975 to 2014 were averaged to create the plot. When compared to the raw GCMs, the bias-corrected GCMs performed better for rainfall, Tmx and Tmn. The model's close resemblance to the CRU data demonstrates the effectiveness of the RF in BC.

4.4.3 Taylor diagram of the bias-corrected GCMs

Figure 11 presents the evaluation of raw and bias-adjusted GCMs in reproducing observed rainfall and temperature using the Taylor diagram. It offers a concise representation of three metrics in a single visual display. The distance between the model and the observation, indicated by the hollow circle on the x-axis, reflects the general level of agreement (error) between the model and the data. The inclination relative to the horizontal axis indicates the degree to which the model accurately


Near
Far
SSP126
SSP245
SSP370
SSP588
Rainfall (%)
20.7
16.7
12.7
8.7
4.7
0.7
-3.3
-7.3
-11.3

◂**Fig. 12** Spatial distribution of changes in rainfall (%) over Nigeria in two future periods for four shared socioeconomic scenarios

represents the magnitude of observed fluctuations, while the correlation along the horizontal axis indicates the extent to which the model replicates the spatial pattern of observed changes. In the figure, triangles represent the raw GCMs, while circles indicate the bias-corrected GCMs. Results demonstrate a significant improvement in the performance of all GCMs after bias correction. The error was reduced significantly, variability became similar to observation, and the spatial correlation improved to above 0.95 for all variables. The figure further reveals that the bias-corrected models cluster closely around the observation point, suggesting a convergence in performance across all GCMs for all variables after bias correction.

4.5 Projection of rainfall and temperatures

Figure 12 depicts the spatial distribution of percentage changes in rainfall expected for Nigeria for the years 2020–2059 (near) and 2060–2099 (far) compared to the historical period (1975–2014). Compared to other SSPs, the percentage change in rainfall for SSP585 was anticipated to be the largest in both the immediate and long term, with increases ranging from 0.7% to 16.7% in the northern region. In the North Eastern portion of the country, a higher percentage increase in rainfall was anticipated in the far future than in the near future in the range of 2.7–12.7% for SSP245, SSP370, and SSP585, but the opposite was true for SSP126, where rainfall changes decreased in the range of 8.7–7.3%. In addition, a projected drop in rainfall change was seen in several southeast sections in the far future of SSP370 and SSP585, ranging from -3.3% to -10%.

Figure 13 displays the spatial distribution pattern of the ensemble model's anticipated increase in Tmx for the near and far futures under the four alternative scenarios. For SSPs, the southern part of the country is expected to experience a temperature rise from 0.70 °C to 1.200 °C in the near future. All four scenarios predicted a more significant increase in Tmx within the northern portions in the long future than in the near future. The Tmx in the northern half of SSP126 was anticipated to increase by 0.41–1.20 °C from the near future to the far future, but SSP245, SSP370, and SSP585 had larger increases in the ranges of 0.21–1.71 °C, 0.21–2.21 °C, and 0.71–2.71 °C, respectively. The greatest temperature increase was recorded in SSP585 for both the immediate and long term, in the ranges of 0.21–1.71 °C and 2.21–2.71 °C, respectively.

Figure 14 depicts the spatial distribution pattern of projected Tmn by the ensemble model for the near and long future under four alternative scenarios. Tmn increased in temperature rapidly across all SSPs, as did Tmx in both the near and long future. In the case of SSP126, the Tmn in the northern section was anticipated to increase by 0.63–1.3 °C from the near future to the distant future. Temperature increases of 1.13 °C–2.13 °C and 1.13–2.63 °C were predicted for SSP245, SSP370. The greatest increase of 2.63 °C and 3.13 °C was predicted for SSP585 in the far future. The estimated increase in Tmn was expected to outnumber the increase in Tmx.

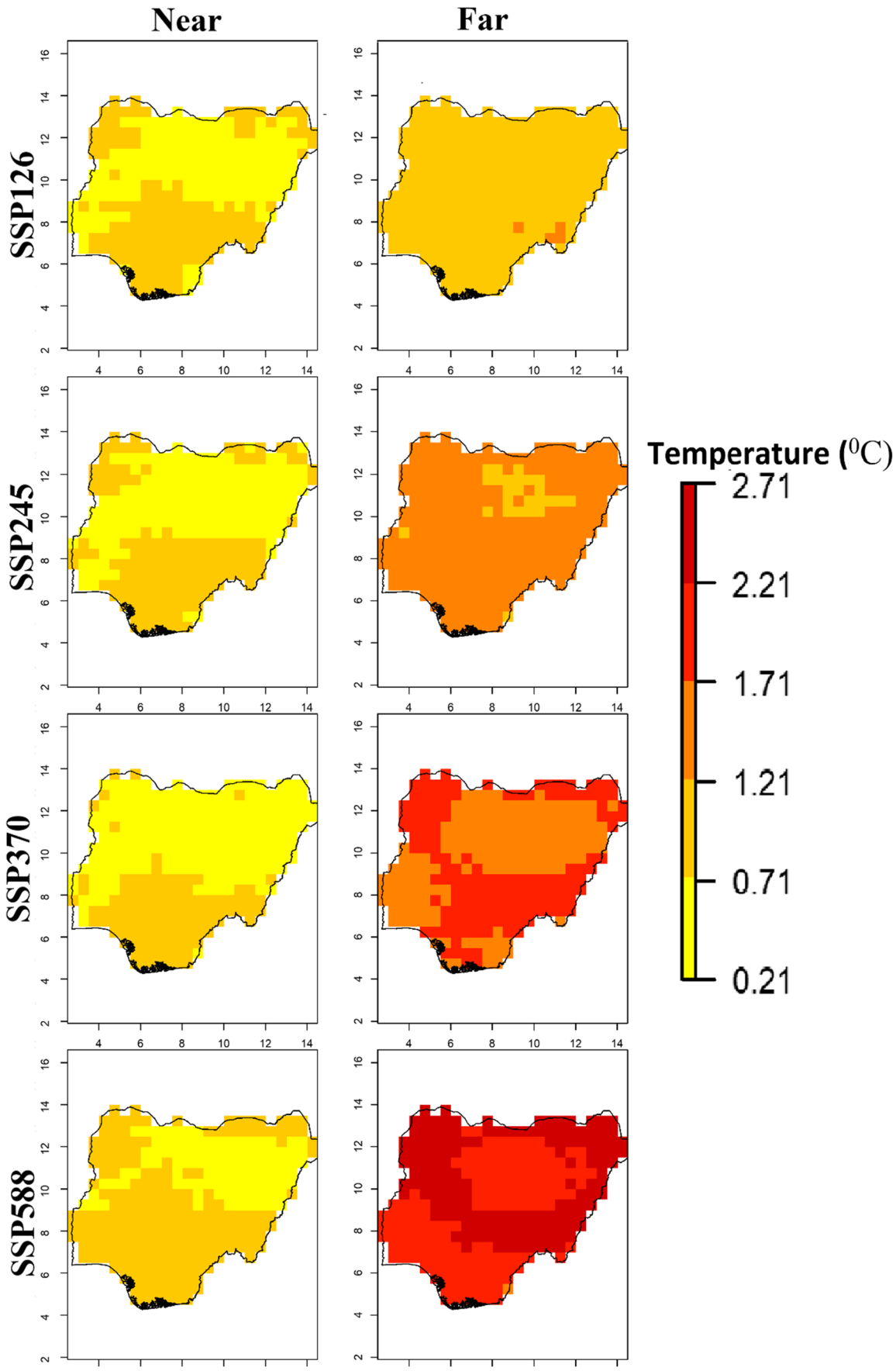
4.6 Probability distribution of the projected rainfall and temperature

Figure 15 depicts a density distribution map of historical and bias-corrected projected rainfall, Tmx, and Tmn from the four GCMs for the four scenarios, namely SSP126, SSP245, SSP370, and SSP585. These charts demonstrate how the distribution of projected and historical data has changed.

For rainfall, all scenarios almost projected the same annual rainfall distribution over the two future eras (near and far). Still, the changes projected by SSP126 and SSP245 in the far future are more significant than those projected by SSP370 and SSP585. The forecasts for high rainfall incidence were noticeably more elevated in the four future scenarios than in the historical under both future periods, implying a rise in less rainy months.

Tmx has shifted in the correct direction for the four future scenarios compared to the historical period in both the near and far futures, indicating a hotter environment. The near-future change was roughly + 1.0 °C, and the far-future shift was about + 1.60 °C. In addition, the projected yearly rainfall distribution for all SSPs in the near future is nearly the same. The opposite is true for the far future, as the average Tmx distribution varies across scenarios. SSP585 is the most common, followed by SSP370, SSP245, and SSP126.

Tmn showed a similar shift to Tmx, indicating a hotter climate prediction in the future, although the change in the near future was predicted to be + 1.2 °C, which was smaller than the shift in the distant future, within + 1.8 °C. Furthermore, compared to the other SSPs and historical data, the SSP585 scenario has the highest occurrence for high-temperature projections in the far future.


Near
Far
SSP126
SSP245
SSP370
SSP588
Temperature (⁰C)
2.71
2.21
1.71
1.21
0.71
0.21

◂**Fig. 13** Spatial distribution of changes in maximum temperature (°C) over Nigeria in two future periods for four shared socioeconomic scenarios

5 Discussion

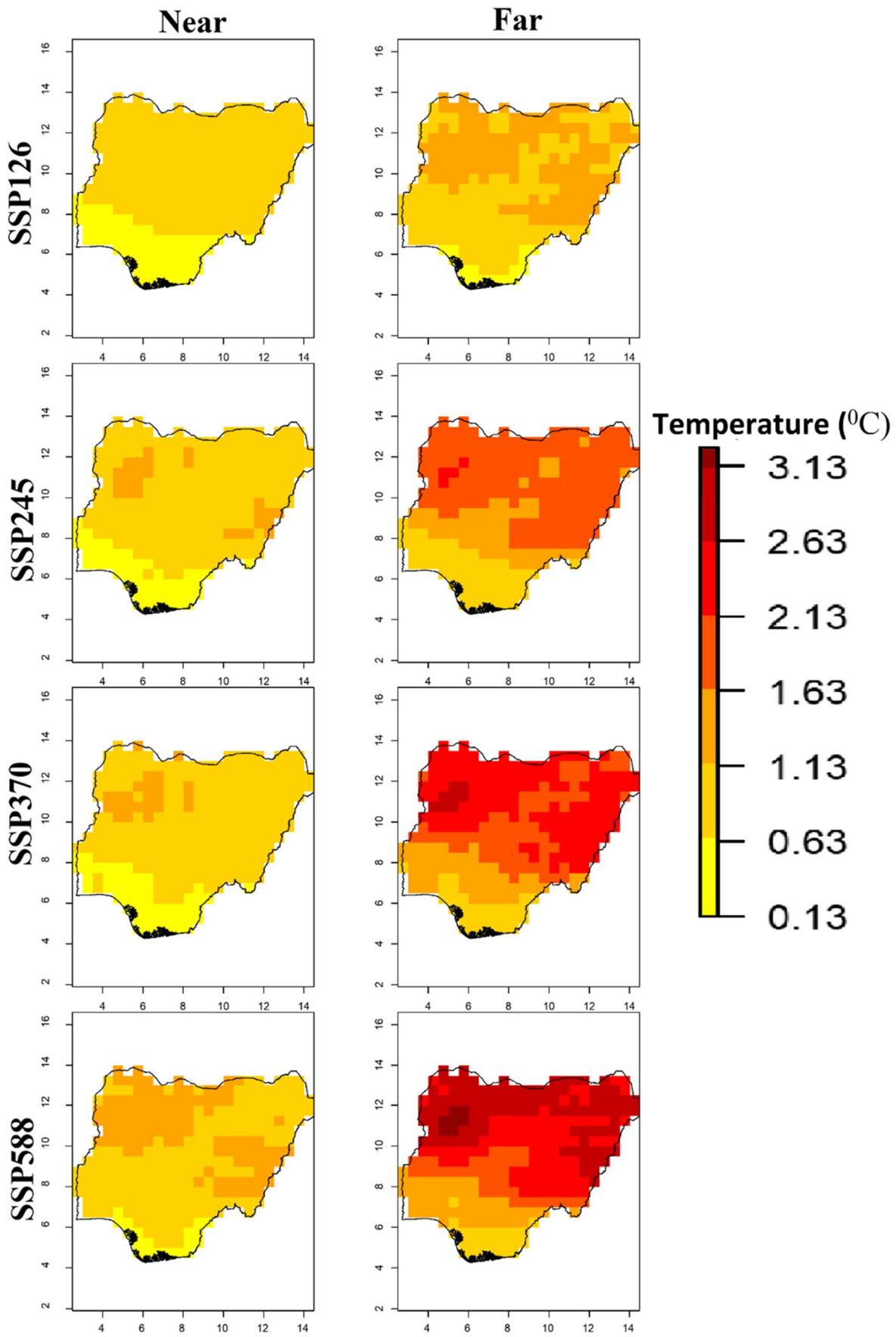
As the effects and possible repercussions of climate change become more evident, it is becoming increasingly vital to give accurate regional estimates for various climate-related elements such as precipitation and temperature. (Dellink et al., 2019; Kopytkovskiy et al., 2015). Water resources and agriculture are critical to life's sustainability, and both are expected to be badly damaged by climate change worldwide. (Sediqi et al., 2019; Shiru, Shahid, et al., 2020). Climate change has caused various problems in Nigeria, including changes in rainfall patterns and rising temperatures. (Durodola, 2019; Okon et al., 2021). These demand improved water resource management, urban development, agriculture, and overall societal well-being (Ngene et al., 2021). GCMs are used to project future climate to strategize policies for societal climate resiliency. Nonetheless, correcting the GCM bias has become critical since there will be systematic mistakes in GCM simulation and future projection without bias correction. (Nashwan et al. 2020; Senatore et al. 2022). This study reveals that raw GCMs overestimate historical rainfall, Tmx and Tmn in Nigeria.

The CPI methodology offers a systematic and adaptable strategy for choosing models, especially when many objectives need to be considered (Ringuest & Ringuest 1992). It recognizes that there may not be ideal solutions simultaneously meeting all objectives. Hence, its mission is to ascertain resolutions that attain a rational middle ground among conflicting goals while considering their significance. Significantly, the CPI categorizes models according to their ability to attain optimal results for all objectives in combination effectively (Salman et al. 2019). An inherent limitation of CPI is that variations in the measurements employed in research might impact the comparability of results. This study tackles this difficulty by standardizing all measurements to a uniform range from 0 to 1. Consequently, the downscaling model's rating in this study may be deemed trustworthy.

This study ranked RF as the best method for reducing bias in climate variables in Nigeria. The results of the present study agree with the findings of Cho et al. (2020) and Feng et al. (2019). Furthermore, in recent years, academics have used machine learning technologies to improve the accuracy of estimates for several high-impact meteorological phenomena. The better performance of RF may be due to its inherent capacity to handle nonlinear relationships (Biau and Scornet 2016; Qi 2012). RF utilizes an ensemble of decision trees where each tree represents a subset of the data and provides unique insights into complicated patterns. The aggregation of predictions from many trees reduces the likelihood of overfitting to noise in the data, resulting in more generalizable models. RF decision trees can represent nonlinear connections without the need for explicit transformation, in contrast to linear models, which rely on assumptions of linearity (Biau and Scornet 2016). Trees partition the data space into smaller, more uniform sections, efficiently capturing nonlinear patterns and relationships among variables. RF algorithm automatically detects the key elements essential for making accurate predictions. It effectively reduces irrelevant information and uncovers the critical variables and relationships. In addition, it trains each tree on a randomly selected sample of data, thus mitigating the influence of outliers and noise on the entire model (Qi 2012). Previous research has also demonstrated the exceptional ability of RF to capture nonlinear connections in climate data (Pour et al. 2016; Lee et al., 2017; Nashwan and Shahid 2019; Yoo et al., 2019).

ML models' performance is intrinsically linked to the attributes of the data they have encountered (Ghahramani 2015; Greener et al. 2022). Climatic patterns, the connections between variables, and the fundamental physical mechanisms can change greatly among various climatic zones. This variability impacts the performance of the model. Hence, a downscaling model that demonstrates exceptional performance in one geographical area may exhibit subpar results in another due to variations in climatic dynamics and data attributes. Different areas may provide varying rankings in terms of accuracy and competence, underscoring the significance of evaluating models in a context-specific manner. Consequently, to guarantee accurate outcomes and well-informed decision-making, it is imperative to thoroughly test downscaling models using data from the precise location where they will be utilized (Najafi et al. 2011; Eum et al. 2017). Furthermore, it is important to note that the validation data's quality and representativeness substantially influence the validation findings' dependability (Sheng et al. 2008). Choosing performance measures to evaluate various facets of model correctness can also substantially impact the ultimate ranking (Muhammad et al. 2019). This work tackled these challenges by utilizing gridded climate data that have undergone quality assurance and employing metrics that assess the fundamental criteria a bias correction model should meet.

This study projected a rainfall change ranging from -11.3% to 20.7% by 2020–2099. Notably, the increase in the northern region is expected to be higher in the near future than in the far future. Conversely, the country's


Near
Far
SSP126
SSP245
SSP370
SSP588
Temperature (0C)
3.13
2.63
2.13
1.63
1.13
0.63
0.13

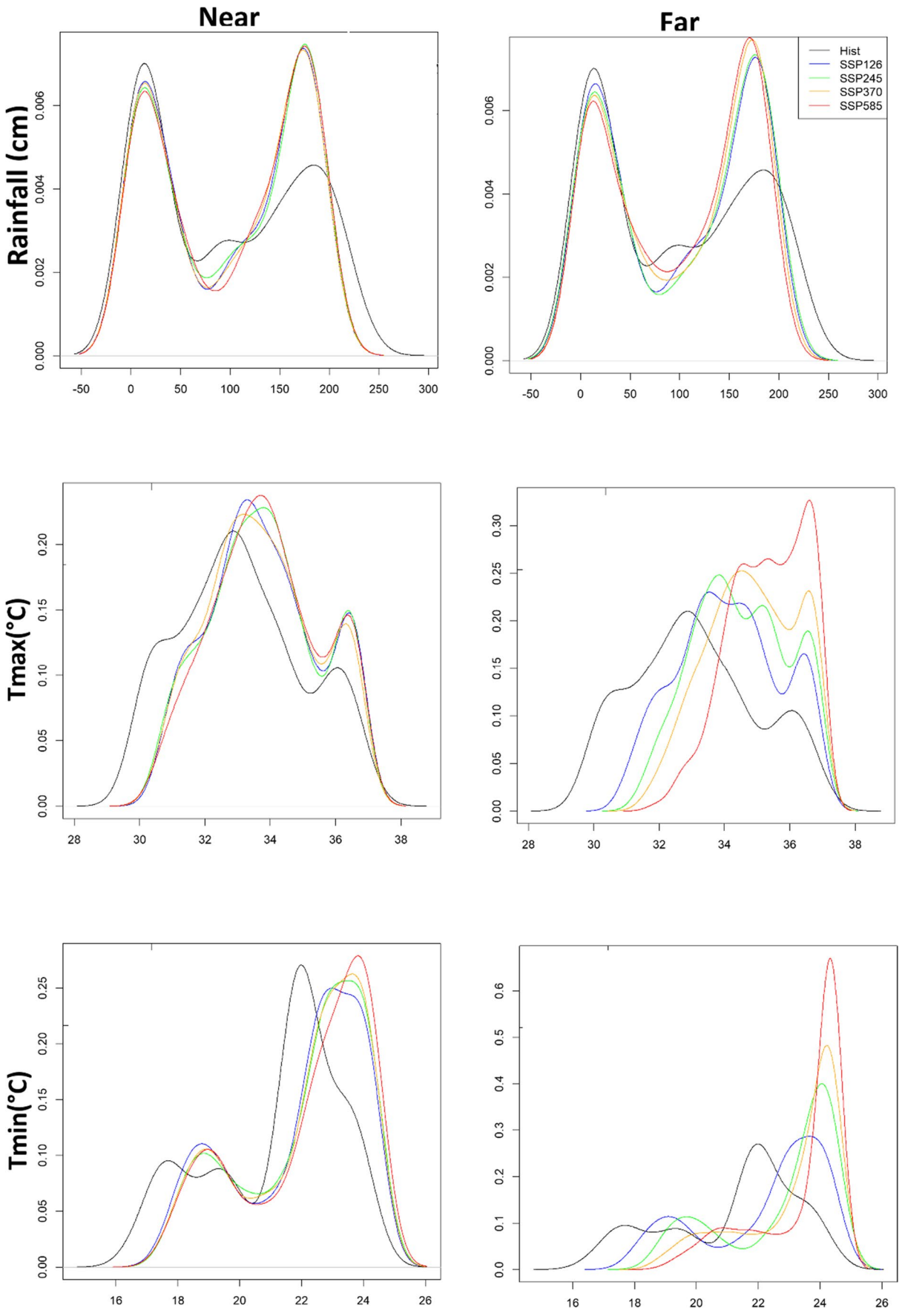
◂**Fig. 14** Spatial distribution of changes in minimum temperature (°C) over Nigeria in two future periods for four shared socioeconomic scenarios

southern half is projected to experience a decrease in rainfall, contributing to the overall variation. This study projected an increase in maximum and minimum temperatures, ranging from 0.21 to 2.71 °C for maximum and 0.13 to 3.13 °C for minimum temperature. Several previous studies have also projected rainfall and temperature for Nigeria using models from different CMIPs. However, their higher northeast temperature increase aligns with this study. Shiru et al. (2019) used CMIP5 GCMs to project a 0–20% rainfall increase in the north and a small decrease in the south. This study also found a similar pattern with up to a 20.7% increase in the north and an 11.3% decrease in the southeast. For temperature, Shiru et al. (2020) employed a multimodel ensemble of bias-corrected CMIP5 GCMs and estimated a maximum temperature increase of 3.0–6.9 °C in the north and 0.0–4.0 °C in the south for RCP 8.5. This study also found a higher temperature increase in the north, but its projected maximum (2.71–3.13 °C) was much lower. Hassan et al. (2020) projected the Niger Delta climate using CMIP6 GCMs under RCP8.5, reporting a 1.38–2.02 °C temperature increase, similar to this study's estimated increase for SSP585 in the same region.

Altered precipitation patterns in Nigeria have the potential to disturb customary periods for planting and harvesting, necessitating farmers to adjust their choice of crops and the timing of their agricultural activities. A decrease in the Diurnal Temperature Range (DTR) results in less cooling throughout the night, which might cause higher evaporation rates and strain water resources used for irrigation (Motha and Baier 2005). Elevated temperature can result in thermal strain, especially in susceptible demographics (Kingsolver et al. 2013). Disruptions in agriculture and frequent extreme temperature events can adversely affect rural lifestyles and economies. The consequences would differ in various parts of Nigeria, contingent upon the projected climatic patterns and ecology. The northern Sudan Savanna zone is expected to have elevated health risks due to heat stress caused by escalating temperatures. However, increased precipitation might mitigate water scarcity and improve agricultural productivity. The variable distribution of rainfall and a substantial increase in temperature in the Guinea Savanna Zone in central-north Nigeria might lead to prolonged droughts and could upset traditional agricultural practices, ultimately leading to crop failures. The decline in rainfall and rise in temperature in the southwest Guinea Savanna Zone might result in changes to vegetation and habitat conditions, posing a threat to endangered species and ecosystems, hence impacting conservation efforts. The combination of decreased precipitation and heightened evaporation due to elevated temperatures in the southeastern Forest zone may pose a significant risk to tropical forests, resulting in the depletion of biodiversity and disturbances in ecosystem functions. Communities relying on forest resources for revenue and sustenance will be affected by alterations in vegetation and the availability of resources. The increased temperature in the southern Mangrove Zone might increase evapotranspiration and potentially disrupt the freshwater flow into the coastal river, leading to a rise in salt levels and increased coastal susceptibility. Nigeria must implement location-specific adaptation strategies to mitigate the effects of climate change. The downscaling model selected in this work can be valuable for accurate downscaling of future climatic conditions at a fine resolution, enabling the generation of pertinent information and determining appropriate adaption strategies.

6 Conclusion

This work addressed a research gap by conducting a comparative analysis of several BC methodologies, including machine learning and conventional statistical methods, for the distinct climatic zones in Nigeria. The objective was to identify the most effective method for reducing the bias in GCM rainfall, Tmx, and Tmn. The study found RF to be the best BC method for correcting GCM biases in Nigeria. RF was then used to adjust the future projections under the SSP126, SSP245, SSP370, and SSP585 for the immediate and long term. According to the study, rainfall is expected to increase in the near future, particularly in the northern region, but to decrease in the long term, with a major decline in Nigeria's southeast and south-south regions. Tmx and Tmn variations are expected to be very comparable.


Near
Far
Rainfall (cm)
Tmax(°C)
Tmin(°C)
Hist
SSP126
SSP245
SSP370
SSP585

◂**Fig. 15** Probability density plots for the historical and projected bias correction rainfall, Tmx and Tmn (rows) under two near and far future periods (columns) for the ensemble GCMs

Tmx and Tmn increases are projected to be greater in the northern region under all four scenarios, with the greatest occurring in the distant future for SSP585, nearly 2.71 °C and 3.13 °C, respectively. This indicates that the DTR is shrinking. The altered rainfall and elevated temperature would disrupt agriculture and increase heat stress conditions, adversely affecting the living standards and economy. The climate projection maps generated in this study provide practical and valuable information for stakeholders in various sectors to effectively plan for and alleviate the adverse impacts of climate change. The downscaled data can also be utilized to adapt to and mitigate climate change. The most optimal BC approach found in this research can be employed to provide high-resolution downscaling of future climate forecasts, hence enhancing comprehension of spatial variability within and among distinct climatic zones. Additional research may be undertaken to create climate impact models that consider these more detailed dynamics at a regional level to evaluate specific vulnerabilities and provide valuable information for focused adaption plans. Further research may also be undertaken to measure the uncertainty in climate projections by various downscaling techniques to choose the most dependable downscaled climate data. Besides, advanced ML models like a deep learning model known as long short-term memory and others can be evaluated in correcting GCMs' biases.

Authors contribution All the authors participated in the conceptualization of the research. B.T., A.D.B. and S.A.A. gathered data. M.K. and M.K.I.M. pre-processed the data. S.S. developed the code for data analysis. B.T., M.A.A. and Z.M.Y. generated results. B.T. and M.K. analyzed the results. B.T. and S.S. prepared the first draft of the article. All the authors contributed to revising and editing the draft. All authors read and approved the final manuscript.

Funding The authors received no financial support for this research, authorship, or publication.

Data availability All the data are available in the public domain at the links provided in the texts.

Code availability The codes used for data processing can be provided on request to the corresponding author.

Declarations

Competing interests The authors declare no competing interests.

References

Adeaga O, Lawal O, Adedeji O, Akinbaloye O (2022) Assessment of vegetation cover dynamics in the agroecological Zones of Nigeria. Bull Geograph Phys Geograph Ser 22:19–32

Ahmad MH, Abubakar A, Ishak MY, Danhassan SS, Jiahua Z, Alatalo JM (2023) Modeling the influence of daily temperature and precipitation extreme indices on vegetation dynamics in Katsina State using statistical downscaling model (SDM). Ecol Ind 155:110979

Ahmed K, Sachindra DA, Shahid S, Iqbal Z, Nawaz N, Khan N (2020) Multimodel ensemble predictions of precipitation and temperature using machine learning algorithms. Atmos Res 236:104806

Ali S, Eum H-I, Cho J, Dan L, Khan F, Dairaku K, Shrestha ML, Hwang S, Nasim W, Khan IA (2019) Assessment of climate extremes in future projections downscaled by multiple statistical downscaling methods over Pakistan. Atmos Res 222:114–133

Amengual A, Homar V, Romero R, Alonso S, Ramis C (2012) A statistical adjustment of regional climate model outputs to local scales: application to Platja de Palma. Spain J Climate 25(3):939–957

Audu HO, Audu NL, Binbol JN, Gana EB (2013) Climate change and its implication on agriculture in Nigeria. Abuja Journal of Geography and Development 3(2):1–15. http://works.bepress.com/abujajournalofgeographyanddevelopment_geographyandenvironmentalmanagementdepartment/8/. Accessed 23 Feb 2024

Ayanlade A, Atai G, Jegede MO (2019) Variability in atmospheric aerosols and effects of humidity, wind and InterTropical discontinuity over different ecological zones in Nigeria. Atmos Environ 201:369–380

Balogun IA, Ntekop AA, Daramola MT (2019) Assessment of the bioclimatic conditions over some selected stations in Nigeria. SN Appl Sci 1:1–14

Beyer R, Krapp M, Manica A (2020) An empirical evaluation of bias correction methods for palaeoclimate simulations. Climate Past 16(4):1493–1508

Biau G, Scornet E (2016) A random forest guided tour. Test 25:197–227

Cannon AJ, Sobie SR, Murdock TQ (2015) Bias correction of GCM precipitation by quantile mapping: how well do methods preserve changes in quantiles and extremes? J Clim 28(17):6938–6959

Casanueva A, Herrera S, Iturbide M, Lange S, Jury M, Dosio A, Maraun D, Gutiérrez JM (2020) Testing bias adjustment methods for regional climate change applications under observational uncertainty and resolution mismatch. Atmos Sci Lett 21(7):e978

Cho D, Yoo C, Im J, Cha D (2020) Comparative assessment of various machine learning-based bias correction methods for numerical weather prediction model forecasts of extreme air temperatures in urban areas. Earth Space Sci 7(4):e2019EA000740

Chokkavarapu, N, Mandla, VR (2019) Comparative study of GCMs, RCMs, downscaling and hydrological models: a review toward future climate change impact estimation. SN Appl Sci, 1(12). https://doi.org/10.1007/s42452-019-1764-x

Cortes C, Vapnik V (1995) Support-Vector Networks. Mach Learn 20:273–297

da Silva LBL, Alencar MH, de Almeida AT (2022) A novel spatiotemporal multi-attribute method for assessing flood risks in urban spaces under climate change and demographic scenarios. Sustain Cities Soc 76:103501

Daniel, H (2023) Performance assessment of bias correction methods using observed and regional climate model data in different watersheds, Ethiopia. J Water Climate Change https://doi.org/10.2166/wcc.2023.115

Dellink R, Lanzi E, Chateau J (2019) The sectoral and regional economic consequences of climate change to 2060. Environ Resour Econ 72:309–363

Dixon KW, Lanzante JR, Nath MJ, Hayhoe K, Stoner A, Radhakrishnan A, Balaji V, Gaitán CF (2016) Evaluating the stationarity assumption in statistically downscaled climate projections: is past performance an indicator of future results? Clim Change 135:395–408

Durodola OS (2019) The impact of climate change induced extreme events on agriculture and food security: a review on Nigeria. Agric Sci 10(4):487–498

Enayati M, Bozorg-Haddad O, Bazrafshan J, Hejabi S, Chu X (2021) Bias correction capabilities of quantile mapping methods for rainfall and temperature variables. J Water Climate Change 12(2):401–419

Enyidi UD (2017) Potable Water and National Water Policy in Nigeria (A historical synthesis, pitfalls and the way forward). J Agricultural Econ Rural Dev 3(2):105–111

Eum HI, Cannon AJ, Murdock TQ (2017) Intercomparison of multiple statistical downscaling methods: multi-criteria model selection for South Korea. Stoch Env Res Risk Assess 31:683–703

Falola T, Heaton MM (2008) A history of Nigeria. Cambridge University Press

Feng R, Zheng HJ, Gao H, Zhang AR, Huang C, Zhang JX, ..., Fan JR (2019) Recurrent Neural Network and random forest for analysis and accurate forecast of atmospheric pollutants: a case study in Hangzhou, China. J Clean Prod 231:1005–1015

Gebresellase SH, Wu Z, Xu H, Muhammad WI (2022) Evaluation and selection of CMIP6 climate models in Upper Awash Basin (UBA), Ethiopia: Evaluation and selection of CMIP6 climate models in Upper Awash Basin (UBA). Ethiopia Theoretical Appl Climatol 149(3–4):1521–1547

Ghahramani Z (2015) Probabilistic machine learning and artificial intelligence. Nature 521(7553):452–459

Gori A, Lin N, Xi D, Emanuel K (2022) Tropical cyclone climatology change greatly exacerbates US extreme rainfall–surge hazard. Nat Clim Chang 12(2):171–178

Greener JG, Kandathil SM, Moffat L, Jones DT (2022) A guide to machine learning for biologists. Nat Rev Mol Cell Biol 23(1):40–55

Gudmundsson L, Bremnes JB, Haugen JE, Skaugen TE (2012) Downscaling RCM precipitation to the station scale using quantile mapping–a comparison of methods. Hydrol Earth Syst Sci Discuss 9(5):6185–6201

Gutjahr O, Heinemann G (2013) Comparing precipitation bias correction methods for high-resolution regional climate simulations using COSMO-CLM: Effects on extreme values and climate change signal. Theoret Appl Climatol 114:511–529

Harris I, Osborn TJ, Jones P, Lister D (2020) Version 4 of the CRU TS monthly high-resolution gridded multivariate climate dataset. Scientific Data 7(1):109

Hartmann, DL (2016) Chapter 11–global climate models. Global physical climatology, 2nd edn. Elsevier, Boston, 325–360. https://doi.org/10.1016/B978-0-12-328531-7.00011-6

Hassan, I, Kalin, RM, White, CJ, Aladejana, JA (2020) Selection of CMIP5 GCM ensemble for the projection of spatio-temporal changes in precipitation and temperature over the Niger Delta, Nigeria. Water (Switzerland), 12(2). https://doi.org/10.3390/w12020385

Haylock MR, Cawley GC, Harpham C, Wilby RL, Goodess CM (2006) Downscaling heavy precipitation over the United Kingdom: a comparison of dynamical and statistical methods and their future scenarios. Int J Climatol A J Royal Meteorologic Soc 26(10):1397–1415

Heo J-H, Ahn H, Shin J-Y, Kjeldsen TR, Jeong C (2019) Probability distributions for a quantile mapping technique for a bias correction of precipitation data: A case study to precipitation data under climate change. Water 11(7):1475

Hieronymus M (2023) A novel machine learning based bias correction method and its application to sea level in an ensemble of downscaled climate projections. No. EGU23-5572. Copernicus Meetings, February 22, 2023, Vienna, Austria.

Homsi R, Shiru MS, Shahid S, Ismail T, Harun SB, Al-Ansari N, Chau K-W, Yaseen ZM (2020) Precipitation projection using a CMIP5 GCM ensemble model: a regional investigation of Syria. Eng Appl Comput Fluid Mech 14(1):90–106

Illangasingha, S, Koike, T, Rasmy, M, Tamakawa, K, Matsuki, H, Selvarajah, H (2023) A holistic approach for using global climate model (GCM) outputs in decision making. J Hydrol, 130213. https://doi.org/10.1016/j.jhydrol.2023.130213

Isa, Z, Sawa, BA, Abdussalam, AF, Ibrahim, M, Babati, AH, Baba, BM, Ugya, AY (2023) Impact of climate change on climate extreme indices in Kaduna River basin, Nigeria. Environ Sci Pollut Res, 0123456789. https://doi.org/10.1007/s11356-023-27821-5

Iseri Y, Diaz AJ, Trinh T, Kavvas ML, Ishida K, Anderson ML, Ohara N, Snider ED (2021) Dynamical downscaling of global reanalysis data for high-resolution spatial modeling of snow accumulation/melting at the central/southern Sierra Nevada watersheds. J Hydrol 598:126445

Islam, SA, Anwar, AHMF, Bari, M (2023) A simple method of bias correction for GCM derived streamflow at catchment scale. Hydrol Sci J, just-accepted. https://doi.org/10.1080/02626667.2023.2218036

Jaiswal R, Mall RK, Singh N, Lakshmi Kumar TV, Niyogi D (2022) Evaluation of bias correction methods for regional climate models: Downscaled rainfall analysis over diverse agroclimatic zones of India. Earth Space Sci 9(2):e2021EA001981

Jose DM, Dwarakish GS (2022) Bias Correction and trend analysis of temperature data by a high-resolution CMIP6 Model over a Tropical River Basin. Asia-Pac J Atmos Sci 58(1):97–115

Jose DM, Vincent AM, Dwarakish GS (2022) Improving multiple model ensemble predictions of daily precipitation and temperature through machine learning techniques. Sci Rep 12(1):1–25. https://doi.org/10.1038/s41598-022-08786-w

Kim D-I, Kwon H-H, Han D (2019) Bias correction of daily precipitation over South Korea from the long-term reanalysis using a composite Gamma-Pareto distribution approach. Hydrol Res 50(4):1138–1161

Kim H, Ham YG, Joo YS, Son SW (2021) Deep learning for bias correction of MJO prediction. Nat Commun 12(1):3087

Kingsolver JG, Diamond SE, Buckley LB (2013) Heat stress and the fitness consequences of climate change for terrestrial ectotherms. Funct Ecol 27(6):1415–1423

Kopytkovskiy M, Geza M, McCray JE (2015) Climate-change impacts on water resources and hydropower potential in the Upper Colorado River Basin. Journal of Hydrology: Regional Studies 3:473–493

Koutsoyiannis D (2006) Nonstationarity versus scaling in hydrology. J Hydrol 324(1–4):239–254

Lafon T, Dadson S, Buys G, Prudhomme C (2013) Bias correction of daily precipitation simulated by a regional climate model: a comparison of methods. Int J Climatol 33(6):1367–1381

Laux P, Rötter RP, Webber H, Dieng D, Rahimi J, Wei J, Faye B, Srivastava AK, Bliefernicht J, Adeyeri O (2021) To bias correct or not to bias correct? An agricultural impact modelers' perspective on regional climate model data. Agric for Meteorol 304:108406

Leander R, Buishand TA (2007) Resampling of regional climate model output for the simulation of extreme river flows. J Hydrol 332(3–4):487–496

Lee S, Kim JC, Jung HS, Lee MJ, Lee S (2017) Spatial prediction of flood susceptibility using randomforest and boosted-tree models in Seoul metropolitan city, Korea. Geomat Nat Haz Risk 8(2):1185–1203

Li X, Babovic V (2019) Multi-site multivariate downscaling of global climate model outputs: an integrated framework combining quantile mapping, stochastic weather generator and Empirical Copula approaches. Clim Dyn 52(9–10):5775–5799

Ling F, Li Y, Luo J-J, Zhong X, Wang Z (2022) Two deep learning-based bias-correction pathways improve summer precipitation prediction over China. Environ Res Lett 17(12):124025

Luo M, Liu T, Meng F, Duan Y, Frankl A, Bao A, De Maeyer P (2018) Comparing bias correction methods used in downscaling precipitation and temperature from regional climate models: a case study from the Kaidu River Basin in Western China. Water 10(8):1046

Madaki MY, Muench S, Kaechele H, Bavorova M (2023) Climate Change Knowledge and Perception among Farming Households in Nigeria. Climate 11(6):115

Maraun D (2013) Bias correction, quantile mapping, and downscaling: Revisiting the inflation issue. J Clim 26(6):2137–2143

Maraun D (2016) Bias correcting climate change simulations-a critical review. Curr Climate Change Rep 2:211–220

Maraun D, Widmann M (2018) Statistical downscaling and bias correction for climate research. Cambridge University Press

Maraun D, Shepherd TG, Widmann M, Zappa G, Walton D, Gutiérrez JM, Hagemann S, Richter I, Soares PMM, Hall A (2017) Towards process-informed bias correction of climate change simulations. Nat Clim Chang 7(11):764–773

Maraun D, Huth R, Gutiérrez JM, Martín DS, Dubrovsky M, Fischer A, Hertig E, Soares PMM, Bartholy J, Pongrácz R (2019) The VALUE perfect predictor experiment: evaluation of temporal variability. Int J Climatol 39(9):3786–3818

Maraun D, Wetterhall F, Ireson AM, Chandler RE, Kendon EJ, Widmann M, Brienen S, Rust HW, Sauter T, Themeßl M (2010) Precipitation downscaling under climate change: Recent developments to bridge the gap between dynamical models and the end user. Reviews of geophysics, 48(3)

Mendez M, Maathuis B, Hein-Griggs D, Alvarado-Gamboa L-F (2020) Performance evaluation of bias correction methods for climate change monthly precipitation projections over Costa Rica. Water 12(2):482

Mobolade, TD, Pourvahidi, P (2020) Bioclimatic approach for climate classification of Nigeria. Sustainability (Switzerland), 12(10). https://doi.org/10.3390/su12104192

Motha RP, Baier W (2005) Impacts of present and future climate change and climate variability on agriculture in the temperate regions: North America. Clim Change 70(1–2):137–164

Muhammad, MKI, Nashwan, MS, Shahid, S, Ismail, T Bin, Song, YH, Chung, ES (2019) Evaluation of empirical reference evapotranspiration models using compromise programming: A case study of Peninsular Malaysia. Sustainability (Switzerland), 11(16). https://doi.org/10.3390/su11164267

Najafi MR, Moradkhani H, Wherry SA (2011) Statistical downscaling of precipitation using machine learning with optimal predictor selection. J Hydrol Eng 16(8):650–664

Nashwan MS, Shahid S (2019) Symmetrical uncertainty and random forest for the evaluation of gridded precipitation and temperature data. Atmos Res 230:104632

Nashwan MS, Shahid S, Chung E-S (2020) High-resolution climate projections for a densely populated Mediterranean region. Sustainability 12(9):3684

Ngene BU, Nwafor CO, Bamigboye GO, Ogbiye AS, Ogundare JO, Akpan VE (2021) Assessment of water resources development and exploitation in Nigeria: A review of integrated water resources management approach. Heliyon 7(1):e05955. https://doi.org/10.1016/j.heliyon.2021.e05955

Nourani V, Razzaghzadeh Z, Baghanam AH, Molajou A (2019) ANN-based statistical downscaling of climatic parameters using decision tree predictor screening method. Theoret Appl Climatol 137:1729–1746

O'Neill BC, Tebaldi C, Van Vuuren DP, Eyring V, Friedlingstein P, Hurtt G, Knutti R, Kriegler E, Lamarque JF, Lowe J, Meehl GA, Moss R, Riahi K, Sanderson BM (2016) The Scenario Model Intercomparison Project (ScenarioMIP) for CMIP6. Geosci Model Dev 9(9):3461–3482. https://doi.org/10.5194/gmd-9-3461-2016

Oguntunde PG, Abiodun BJ (2013) The impact of climate change on the Niger River Basin hydroclimatology, West Africa. Clim Dyn 40:81–94

Okon EM, Falana BM, Solaja SO, Yakubu SO, Alabi OO, Okikiola BT, ..., Edeme AB (2021) Systematic review of climate change impact research in Nigeria: implication for sustainable development. Heliyon, 7(9):e07941. https://doi.org/10.1016/j.heliyon.2021.e07941

Okwu, MO, Tartibu, LK, Okwu, MO, Tartibu, LK (2021) Artificial neural network. Metaheuristic optimization: nature-inspired algorithms swarm and computational intelligence, theory and applications, 133–145. https://doi.org/10.1007/978-3-030-61111-8_14

Passow C, Donner RV (2020) Regression-based distribution mapping for bias correction of climate model outputs using linear quantile regression. Stoch Env Res Risk Assess 34:87–102

Peng Y, Duan A, Hu W, Tang B, Li X, Yang X (2022) Observational constraint on the future projection of temperature in winter over the Tibetan Plateau in CMIP6 models. Environ Res Lett 17(3):34023

Piani C, Weedon GP, Best M, Gomes SM, Viterbo P, Hagemann S, Haerter JO (2010) Statistical bias correction of global simulated daily precipitation and temperature for the application of hydrological models. J Hydrol 395(3–4):199–215

Pour SH, Shahid S, Chung ES (2016) A hybrid model for statistical downscaling of daily rainfall. Procedia Eng 154:1424–1430

Pour SH, Shahid S, Chung ES, Wang XJ (2018) Model output statistics downscaling using support vector machine for the projection of spatial and temporal changes in rainfall of Bangladesh. Atmos Res 213:149–162

Qi Y (2012) Random forest for bioinformatics. Ensemble Mach Learn: Methods Appl, pp 307–323

Raczko E, Zagajewski B (2017) Comparison of support vector machine, random forest and neural network classifiers for tree species classification on airborne hyperspectral APEX images. Eur J Remote Sens 50(1):144–154

Raju KS, Kumar DN (2014) Ranking of global climate models for India using multicriterion analysis. Climate Res 60(2):103–117. https://doi.org/10.3354/cr01222

Riahi K, Van Vuuren DP, Kriegler E, Edmonds J, O'neill BC, Fujimori S, Bauer N, Calvin K, Dellink R, Fricko O (2017) The Shared Socioeconomic Pathways and their energy, land use, and greenhouse gas emissions implications: An overview. Glob Environ Chang 42:153–168

Ringuest, JL, Ringuest, JL (1992) Compromise programming. Multiobjective optimization: behavioral and computational considerations, 51–59. https://doi.org/10.1007/978-1-4615-3612-3_4

Ruffault J, Martin-StPaul NK, Duffet C, Goge F, Mouillot F (2014) Projecting future drought in Mediterranean forests: bias correction of climate models matters! Theoret Appl Climatol 117:113–122

Salman SA, Shahid S, Ismail T, Al-Abadi AM, Wang XJ, Chung ES (2019) Selection of gridded precipitation data for Iraq using compromise programming. Meas J Int Meas Confederation 132:87–98. https://doi.org/10.1016/j.measurement.2018.09.047

Salman, SA, Hamed, MM, Shahid, S, Ahmed, K, Sharafati, A, Asaduzzaman, M, Ziarh, GF, Ismail, T, Chung, E, Wang, X (2022)

Projecting spatiotemporal changes of precipitation and temperature in Iraq for different shared socioeconomic pathways with selected Coupled Model Intercomparison Project Phase 6. Int J Climatol, 42(16), 9032–9050. https://doi.org/10.1002/joc.7794

Sarr MA, Seidou O, Tramblay Y, El Adlouni S (2015) Comparison of downscaling methods for mean and extreme precipitation in Senegal. J Hydrol : Region Stud 4:369–385

Sediqi MN, Shiru MS, Nashwan MS, Ali R, Abubaker S, Wang X, Manawi SMA (2019) Spatio-temporal pattern in the changes in availability and sustainability of water resources in Afghanistan. Sustainability 11(20):5836

Senatore A, Fuoco D, Maiolo M, Mendicino G, Smiatek G, Kunstmann H (2022) Evaluating the uncertainty of climate model structure and bias correction on the hydrological impact of projected climate change in a Mediterranean catchment. J Hydrol: Region Stud 42:101120

Seo G-Y, Ahn J-B (2023) Comparison of Bias Correction Methods for Summertime Daily Rainfall in South Korea Using Quantile Mapping and Machine Learning Model. Atmosphere 14(7):1057

Sheng, VS, Provost, F, Ipeirotis, PG (2008) Get another label? improving data quality and data mining using multiple, noisy labelers. In Proceedings of the 14th ACM SIGKDD international conference on Knowledge discovery and data mining, Vegas, Nevada, USA, pp. 614–622

Shiru MS, Park I (2020) Comparison of ensembles projections of rainfall from four bias correction methods over Nigeria. Water 12(11):3044

Shiru MS, Shahid S, Chung ES, Alias N (2019) Changing characteristics of meteorological droughts in Nigeria during 1901–2010. Atmos Res 223(March):60–73. https://doi.org/10.1016/j.atmosres.2019.03.010

Shiru MS, Chung E-S, Shahid S, Alias N (2020) GCM selection and temperature projection of Nigeria under different RCPs of the CMIP5 GCMS. Theoret Appl Climatol 141:1611–1627

Shrestha SG, Pradhanang SM (2022) Optimal selection of representative climate models and statistical downscaling for climate change impact studies: a case study of Rhode Island, USA. Theoret Appl Climatol 149(1–2):695–708

Solarin SA, Nathaniel SP, Bekun FV, Okunola AM, Alhassan A (2021) Towards achieving environmental sustainability: environmental quality versus economic growth in a developing economy on ecological footprint via dynamic simulations of ARDL. Environ Sci Pollut Res 28:17942–17959

Su F, Duan X, Chen D, Hao Z, Cuo L (2013) Evaluation of the global climate models in the CMIP5 over the Tibetan Plateau. J Clim 26(10):3187–3208. https://doi.org/10.1175/JCLI-D-12-00321.1

Switanek MB, Troch PA, Castro CL, Leuprecht A, Chang H-I, Mukherjee R, Demaria E (2017) Scaled distribution mapping: a bias correction method that preserves raw climate model projected changes. Hydrol Earth Syst Sci 21(6):2649–2666

Tan Y, Guzman SM, Dong Z, Tan L (2020) Selection of effective GCM bias correction methods and evaluation of hydrological response under future climate scenarios. Climate 8(10):108

Teutschbein C, Seibert J (2012) Bias correction of regional climate model simulations for hydrological climate-change impact studies: Review and evaluation of different methods. J Hydrol 456:12–29

Tumsa BC (2022) Performance assessment of six bias correction methods using observed and RCM data at upper Awash basin, Oromia, Ethiopia. J Water Climate Change 13(2):664–683

Vogel, E, Johnson, F, Marshall, L, Bende-Michl, U, Wilson, L, Peter, JR, Wasko, C, Srikanthan, S, Sharples, W, Dowdy, A (2023) An evaluation framework for downscaling and bias correction in climate change impact studies. J Hydrol, 129693. https://doi.org/10.1016/j.jhydrol.2023.129693

Vrac M, Drobinski P, Merlo A, Herrmann M, Lavaysse C, Li L, Somot S (2012) Dynamical and statistical downscaling of the French Mediterranean climate: uncertainty assessment. Nat Hazard 12(9):2769–2784

Wada, IM, Usman, HS, Nwankwegu, AS, Usman, MN, Gebresellase, SH (2023) Selection and downscaling of CMIP6 climate models in Northern Nigeria. Theoretical and Applied Climatology, 1–19. https://doi.org/10.1007/s00704-023-04534-w

Wilcke RAI, Mendlik T, Gobiet A (2013) Multi-variable error correction of regional climate models. Clim Change 120:871–887

Worako AW, Haile AT, Taye MT (2022) Implication of bias correction on climate change impact projection of surface water resources in the Gidabo sub-basin, Southern Ethiopia. J Water Climate Change 13(5):2070–2088

Wörner V, Kreye P, Meon G (2019) Effects of bias-correcting climate model data on the projection of future changes in high flows. Hydrology 6(2):46

Wu Y, Miao C, Fan X, Gou J, Zhang Q, Zheng H (2022) Quantifying the uncertainty sources of future climate projections and narrowing uncertainties with bias correction techniques. Earth's Future 10(11):e2022EF002963

Xu Z, Han Y, Yang Z (2019) Dynamical downscaling of regional climate: A review of methods and limitations. Sci China Earth Sci 62:365–375

Yoo C, Han D, Im J, Bechtel B (2019) Comparison between convolutional neural networks and random forest for local climate zone classification in mega urban areas using Landsat images. ISPRS J Photogramm Remote Sens 157:155–170

Zhang Q, Li YP, Huang GH, Wang H, Li YF, Liu YR, Shen ZY (2022) A novel statistical downscaling approach for analyzing daily precipitation and extremes under the impact of climate change: Application to an arid region. J Hydrol 615:128730

Publisher's Note Springer Nature remains neutral with regard to jurisdictional claims in published maps and institutional affiliations.

Springer Nature or its licensor (e.g. a society or other partner) holds exclusive rights to this article under a publishing agreement with the author(s) or other rightsholder(s); author self-archiving of the accepted manuscript version of this article is solely governed by the terms of such publishing agreement and applicable law.

View publication stats